

Supplementary Materials

Emergent Intelligence: Resonant Oscillators Produce Proactive Adaptive Behavior

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This document contains supplementary materials for the main article. Contents: Table S1 (threshold sensitivity analysis), Figure S1 (convergence analysis), per-hypothesis distribution statistics and violin plots (Tables S2–S12, Figures S2–S12), external baseline effect sizes (Table S13), and formal statistical tests with Holm-Bonferroni correction (Table S14).

Table S1: Threshold Sensitivity Analysis

For each hypothesis that uses a threshold-based validation check, the table below shows whether the check passes (+) or fails (–) at each threshold value. The stable range indicates the interval over which the conclusion is invariant to threshold choice.

H0

circle_fit_score thresholds:

Check	Value	0.1	0.2	0.3	0.4	0.5
Hob circle_fit > T <i>5/5 pass, stable [0.1, 0.5]</i>	0.6409	+	+	+	+	+
Hoc circle_fit > T <i>5/5 pass, stable [0.1, 0.5]</i>	0.6419	+	+	+	+	+
Hof circle_fit > T <i>5/5 pass, stable [0.1, 0.5]</i>	0.6374	+	+	+	+	+
Hog circle_fit > T <i>5/5 pass, stable [0.1, 0.5]</i>	0.6348	+	+	+	+	+

H1

spiral_quality thresholds:

Check	Value	0.05	0.08	0.1	0.12	0.15	0.18	0.2	0.25	0.3	0.35
H1a spiral_quality > T <i>5/10 pass, stable [0.05, 0.15]</i>	0.1650	+	+	+	+	+	-	-	-	-	-

H2

spiral_quality thresholds:

Check	Value	0.05	0.08	0.1	0.12	0.15	0.18	0.2	0.25	0.3	0.35
H2a stochastic < T <i>10/10 pass, stable [0.05, 0.35]</i>	0.0384	+	+	+	+	+	+	+	+	+	+
H2b inverter < T <i>9/10 pass, stable [0.08, 0.35]</i>	0.0672	-	+	+	+	+	+	+	+	+	+
H2c composite (no cross) < T <i>6/10 pass, stable [0.15, 0.35]</i>	0.1271	-	-	-	-	+	+	+	+	+	+
H2d composite (no stagger) < T <i>9/10 pass, stable [0.08, 0.35]</i>	0.0597	-	+	+	+	+	+	+	+	+	+
H2e opposition only < T <i>9/10 pass, stable [0.08, 0.35]</i>	0.0730	-	+	+	+	+	+	+	+	+	+
H2f staggering only < T <i>8/10 pass, stable [0.1, 0.35]</i>	0.0881	-	-	+	+	+	+	+	+	+	+
H2g both > T <i>10/10 pass, stable [0.05, 0.35]</i>	0.3947	+	+	+	+	+	+	+	+	+	+

H4B

spiral_quality thresholds:

Check	Value	0.05	0.08	0.1	0.12	0.15	0.18	0.2	0.25	0.3	0.35
H4b1 baseline > T <i>7/10 pass, stable [0.05, 0.2]</i>	0.2369	+	+	+	+	+	+	+	-	-	-

H7

spiral_quality thresholds:

Check	Value	0.05	0.08	0.1	0.12	0.15	0.18	0.2	0.25	0.3	0.35
H7a1 D_ROT=0 > T <i>10/10 pass, stable [0.05, 0.35]</i>	0.6521	+	+	+	+	+	+	+	+	+	+
H7a4 D_ROT=0.1 > T <i>9/10 pass, stable [0.05, 0.3]</i>	0.3058	+	+	+	+	+	+	+	+	+	-
H7a6 D_ROT=0.5 > T <i>7/10 pass, stable [0.05, 0.2]</i>	0.2380	+	+	+	+	+	+	+	-	-	-

H8

spiral_quality thresholds:

Check	Value	0.05	0.08	0.1	0.12	0.15	0.18	0.2	0.25	0.3	0.35
H8a1 3-inv > T <i>8/10 pass, stable [0.05, 0.25]</i>	0.2703	+	+	+	+	+	+	+	+	-	-
H8a5 7-inv > T <i>10/10 pass, stable [0.05, 0.35]</i>	0.7196	+	+	+	+	+	+	+	+	+	+

H9

spiral_quality thresholds:

Check	Value	0.05	0.08	0.1	0.12	0.15	0.18	0.2	0.25	0.3	0.35
NEG-H1 stochastic < T <i>9/10 pass, stable [0.08, 0.35]</i>	0.0654	-	+	+	+	+	+	+	+	+	+
NEG-H2 same-C1 < T <i>7/10 pass, stable [0.12, 0.35]</i>	0.1069	-	-	-	+	+	+	+	+	+	+
NEG-H4a all-normal < T	0.0961	-	-	+	+	+	+	+	+	+	+

8/10 pass, stable [0.1, 0.35]

SUMMARY

PARTIAL: h1 / H1a spiral_quality > T: 5/10 thresholds pass
PARTIAL: h2 / H2b inverter < T: 9/10 thresholds pass
PARTIAL: h2 / H2c composite (no cross) < T: 6/10 thresholds pass
PARTIAL: h2 / H2d composite (no stagger) < T: 9/10 thresholds pass
PARTIAL: h2 / H2e opposition only < T: 9/10 thresholds pass
PARTIAL: h2 / H2f staggering only < T: 8/10 thresholds pass
PARTIAL: h4b / H4b1 baseline > T: 7/10 thresholds pass
PARTIAL: h7 / H7a4 D_ROT=0.1 > T: 9/10 thresholds pass
PARTIAL: h7 / H7a6 D_ROT=0.5 > T: 7/10 thresholds pass
PARTIAL: h8 / H8a1 3-inv > T: 8/10 thresholds pass
PARTIAL: h9 / NEG-H1 stochastic < T: 9/10 thresholds pass
PARTIAL: h9 / NEG-H2 same-C1 < T: 7/10 thresholds pass
PARTIAL: h9 / NEG-H4a all-normal < T: 8/10 thresholds pass

Table S1. Threshold sensitivity analysis. Each check is evaluated across threshold values from 0.05 to 0.50. '+' = pass, '-' = fail. Per-check stable ranges are reported; narrowest is [0.05, 0.15] (H1a).

Figure S1: Convergence Analysis

Standard error of the mean (SE) and 95% confidence interval half-width as a function of sample size N for two primary estimands: spiral quality (H2g, positive control) and food eaten (H3a, mode switching). Both metrics follow the expected $1/\sqrt{N}$ decay. At $N = 1,000$, CI half-width is less than 5% of the mean for both estimands.

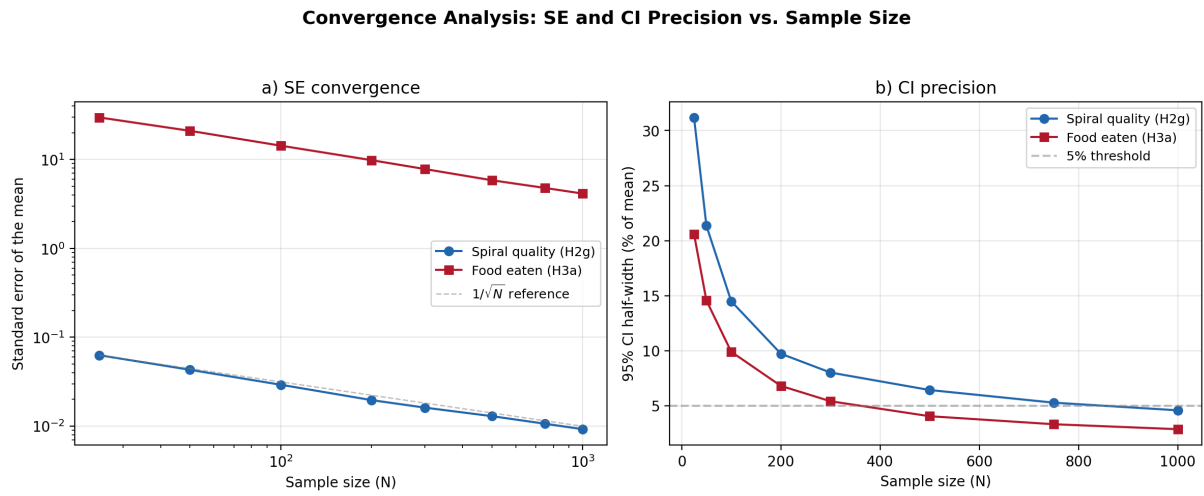


Figure S1. Convergence analysis. (a) SE vs. sample size on log-log scale; dashed line shows the theoretical $1/\sqrt{N}$ reference. (b) 95% CI half-width as percentage of the mean; dashed line marks the 5% threshold. Both estimands converge below 5% well before $N = 1,000$.

Table S2 / Figure S2: H0: Baseline Dynamics

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h0a_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0612	0.0169	0.0882	0.0949	1.94	0.0000	0.5834
area_cells_visited	1000	124.7760	125.0000	4.6339	6.0000	-0.62	105.0000	137.0000
mean_speed	1000	146.3241	142.9991	31.3416	46.6273	1.32	96.3725	312.9344

[h0b_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0729	0.0331	0.0972	0.0954	2.21	0.0000	0.7059
area_cells_visited	1000	25.0690	25.0000	4.0962	6.0000	0.58	16.0000	41.0000
mean_speed	1000	70.0970	70.0970	0.0003	0.0004	-0.04	70.0958	70.0979

[h0c_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0723	0.0293	0.1034	0.0912	2.56	0.0000	0.7565
area_cells_visited	1000	24.8840	25.0000	3.9487	5.0000	0.56	16.0000	44.0000
mean_speed	1000	70.0970	70.0970	0.0003	0.0004	-0.03	70.0959	70.0981

[h0d_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0619	0.0283	0.0831	0.0780	2.26	0.0000	0.6449
area_cells_visited	1000	27.5750	27.0000	4.2882	5.0000	0.45	17.0000	42.0000
mean_speed	1000	140.5063	140.5063	0.0005	0.0007	-0.01	140.5046	140.5080

[h0e_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0888	0.0469	0.1079	0.1181	1.78	0.0000	0.6581
area_cells_visited	1000	31.2720	31.0000	4.7685	6.0000	0.57	20.0000	49.0000
mean_speed	1000	281.3251	281.3251	0.0010	0.0013	-0.11	281.3222	281.3282

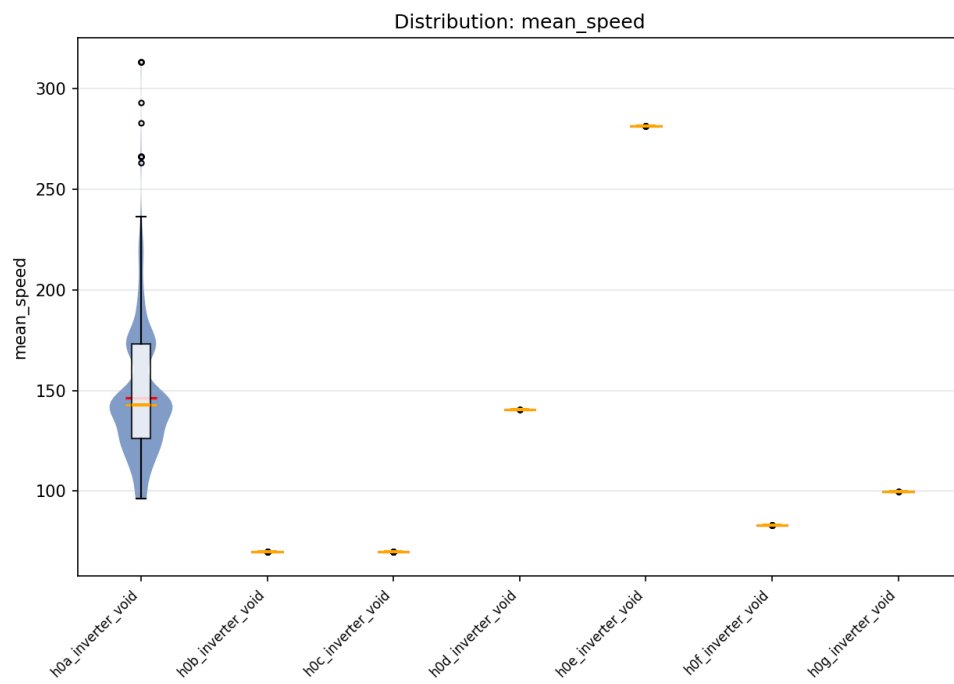
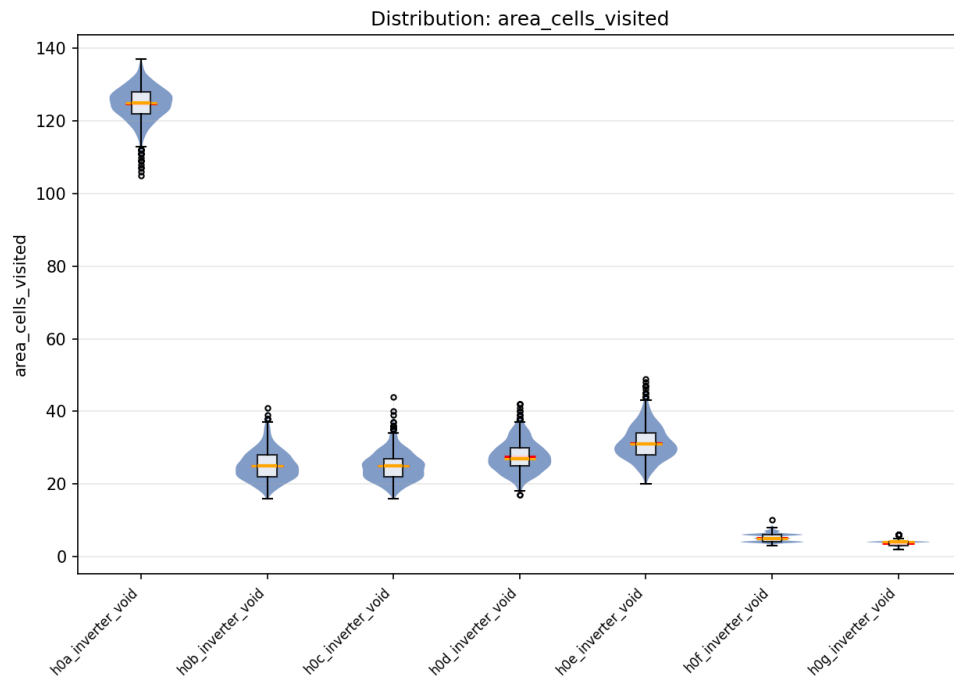
[h0f_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0883	0.0468	0.1057	0.1184	1.72	0.0000	0.6663
area_cells_visited	1000	5.2290	5.0000	1.1062	2.0000	0.41	3.0000	10.0000
mean_speed	1000	83.0095	83.0094	0.0011	0.0014	0.08	83.0055	83.0127

[h0g_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0701	0.0330	0.0916	0.0872	2.13	0.0000	0.6431
area_cells_visited	1000	3.7030	4.0000	0.7781	1.0000	-0.31	2.0000	6.0000
mean_speed	1000	99.6112	99.6112	0.0019	0.0027	0.03	99.6054	99.6169

Table S2. Per-trial distribution statistics for H0: Baseline Dynamics. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.



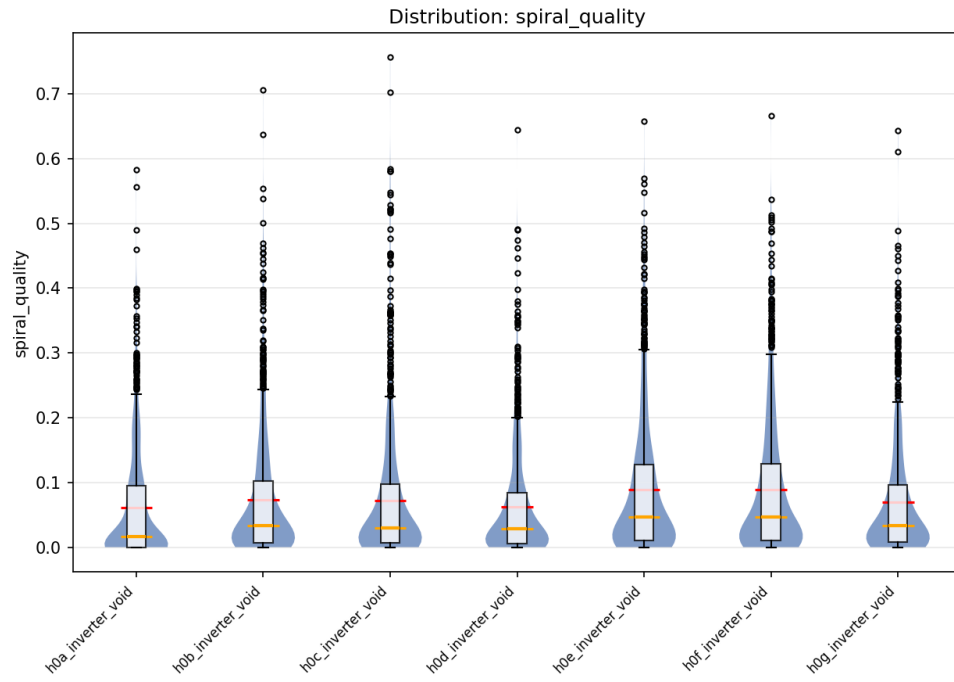


Figure S2. Distribution violin plots for HO: Baseline Dynamics. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S3 / Figure S3: H1: Emergence of Spiral Search

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h1a_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1650	0.0903	0.1867	0.2331	1.43	0.0000	0.9035
area_cells_visited	1000	153.0070	154.0000	11.7961	17.0000	-0.26	112.0000	183.0000
mean_speed	1000	79.1307	77.1547	11.1023	0.0003	7.06	77.1541	234.2948

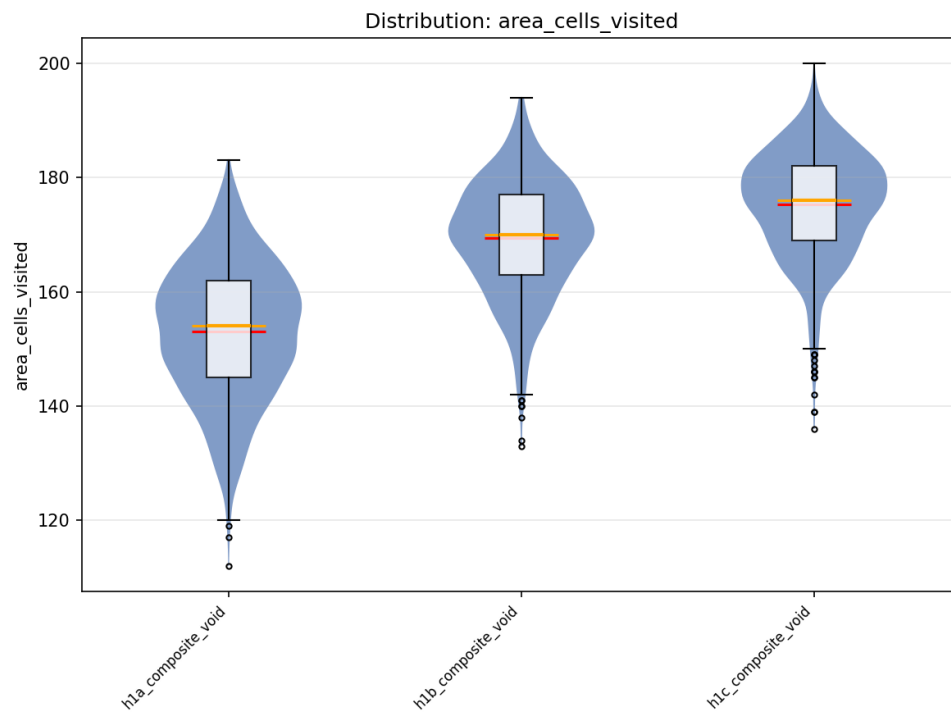
[h1b_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2351	0.1556	0.2410	0.3349	1.04	0.0000	0.9664
area_cells_visited	1000	169.3320	170.0000	10.0451	14.0000	-0.41	133.0000	194.0000
mean_speed	1000	85.1542	78.3034	19.9883	0.0003	3.16	78.3028	213.0123

[h1c_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2899	0.2259	0.2538	0.3992	0.70	0.0000	0.9541
area_cells_visited	1000	175.3370	176.0000	9.3948	13.0000	-0.68	136.0000	200.0000
mean_speed	1000	88.3403	78.9523	22.9709	0.0003	3.22	78.9517	328.5737

Table S3. Per-trial distribution statistics for H1: Emergence of Spiral Search. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.



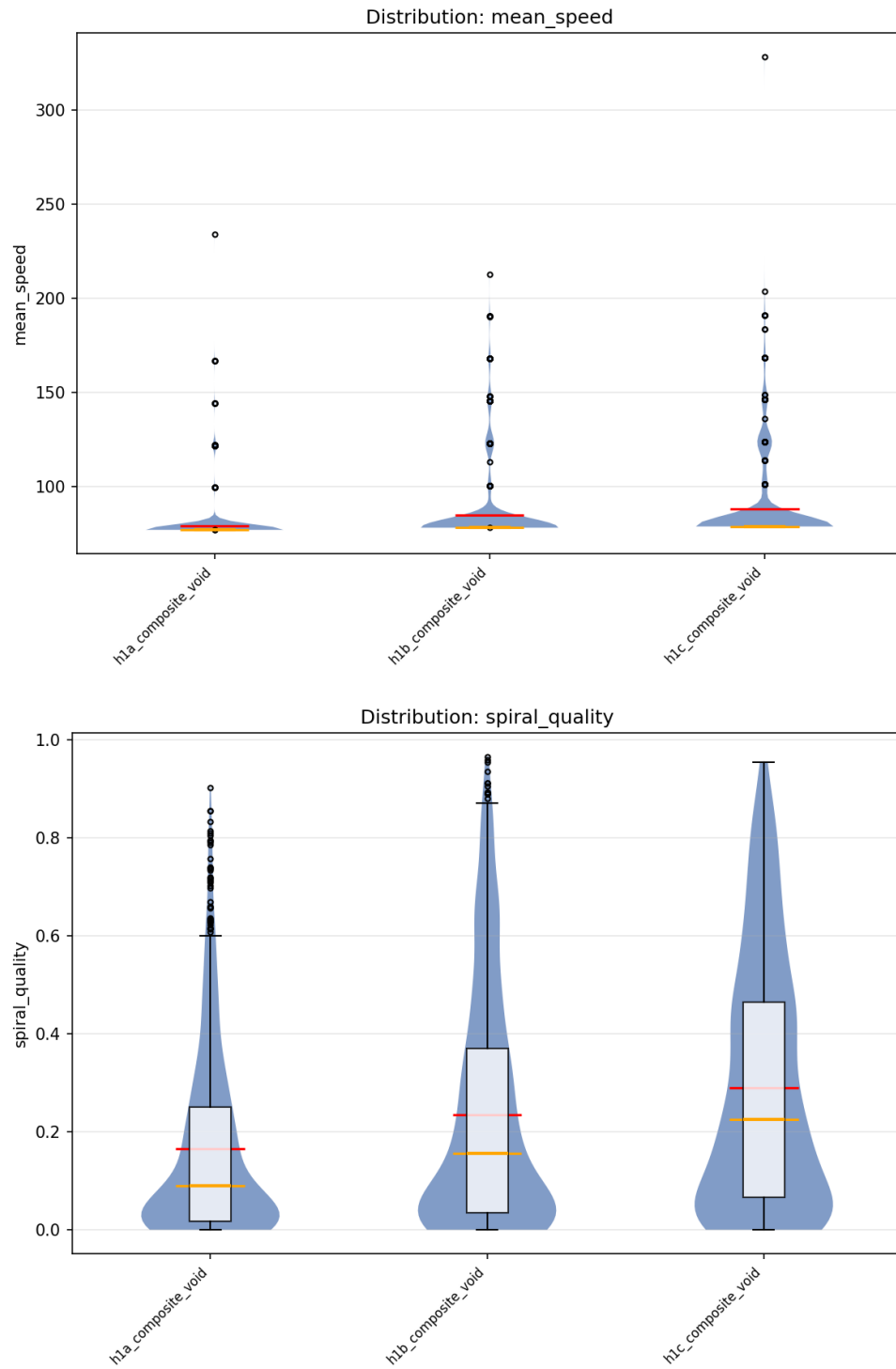


Figure S3. Distribution violin plots for H1: Emergence of Spiral Search. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S4 / Figure S4: H2: Specificity (Ablation)

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h2a_stochastic_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0384	0.0000	0.0961	0.0000	3.02	0.0000	0.6755
area_cells_visited	1000	38.2790	38.0000	5.5021	8.0000	0.16	22.0000	60.0000
mean_speed	1000	39.5752	30.5703	21.1647	8.6899	1.92	18.1079	217.8928

[h2b_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0672	0.0338	0.0914	0.0808	2.52	0.0000	0.6999
area_cells_visited	1000	21.2980	21.0000	3.1974	4.0000	0.46	14.0000	33.0000
mean_speed	1000	69.9599	69.9599	0.0004	0.0005	-0.04	69.9586	69.9613

[h2c_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1271	0.0642	0.1516	0.1733	1.63	0.0000	0.8470
area_cells_visited	1000	49.4090	49.0000	6.8317	9.0000	0.23	32.0000	73.0000
mean_speed	1000	74.2135	74.2135	0.0003	0.0004	-0.00	74.2126	74.2146

[h2d_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0597	0.0276	0.0812	0.0749	2.54	0.0000	0.6415
area_cells_visited	1000	16.1440	16.0000	2.4235	4.0000	0.42	10.0000	24.0000
mean_speed	1000	74.2135	74.2135	0.0005	0.0007	-0.09	74.2119	74.2155

[h2e_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0730	0.0343	0.0962	0.0905	2.30	0.0000	0.7368
area_cells_visited	1000	119.9910	120.0000	16.7432	23.0000	0.14	73.0000	178.0000
mean_speed	1000	209.8787	209.8787	0.0004	0.0006	-0.04	209.8774	209.8800

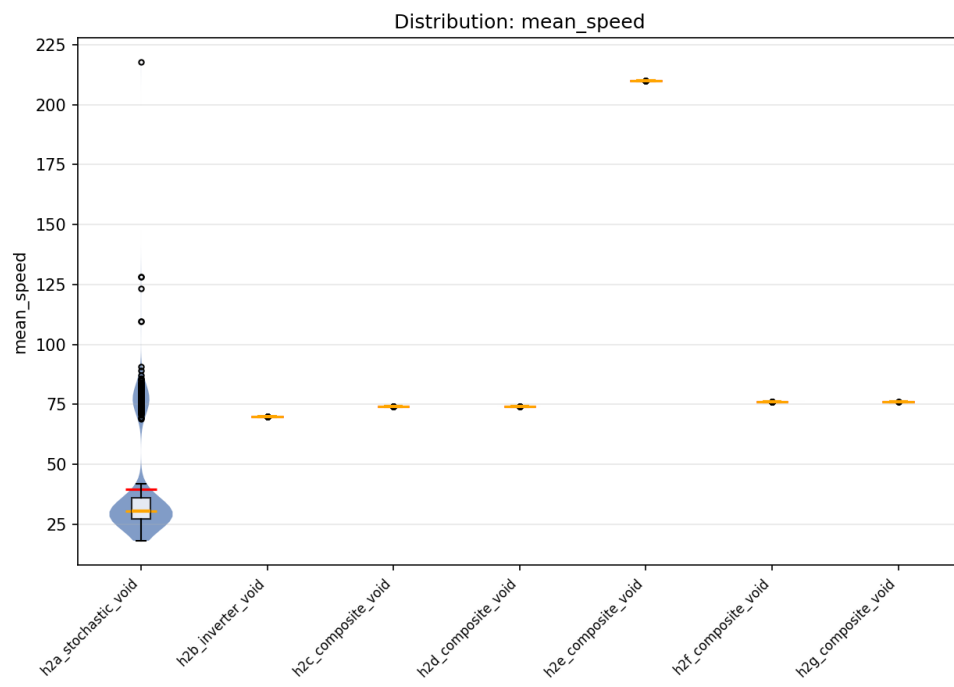
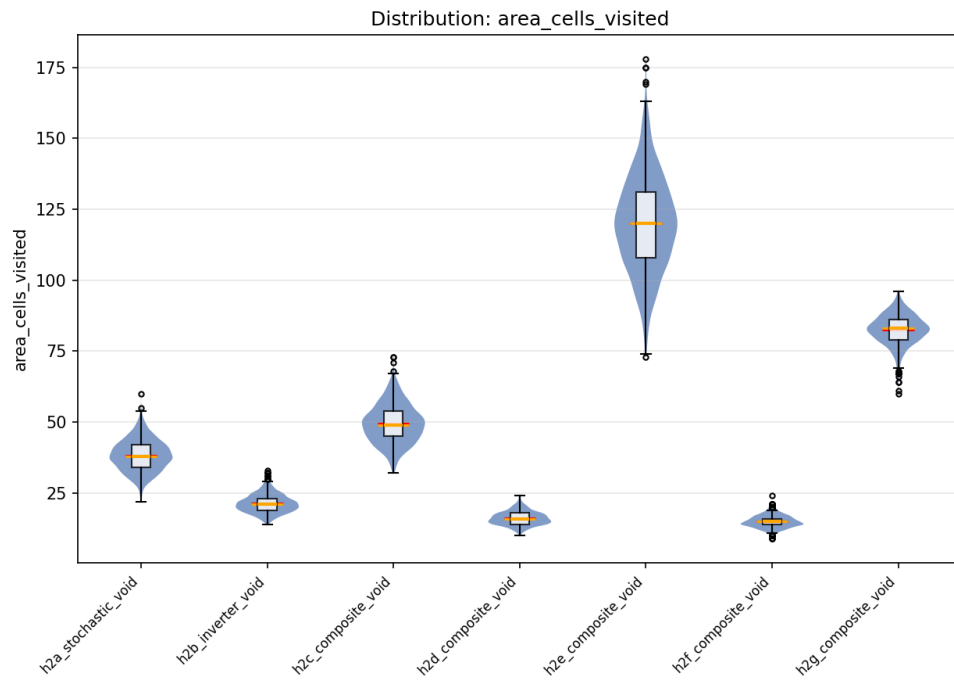
[h2f_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0881	0.0523	0.1034	0.1213	1.82	0.0000	0.6396
area_cells_visited	1000	14.9680	15.0000	2.1849	2.0000	0.32	9.0000	24.0000
mean_speed	1000	76.1598	76.1598	0.0006	0.0007	-0.09	76.1578	76.1614

[h2g_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.3947	0.3737	0.2915	0.5324	0.21	0.0000	0.9934
area_cells_visited	1000	82.6280	83.0000	5.1972	7.0000	-0.45	60.0000	96.0000
mean_speed	1000	76.1598	76.1598	0.0003	0.0004	-0.15	76.1588	76.1607

Table S4. Per-trial distribution statistics for H2: Specificity (Ablation). $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.



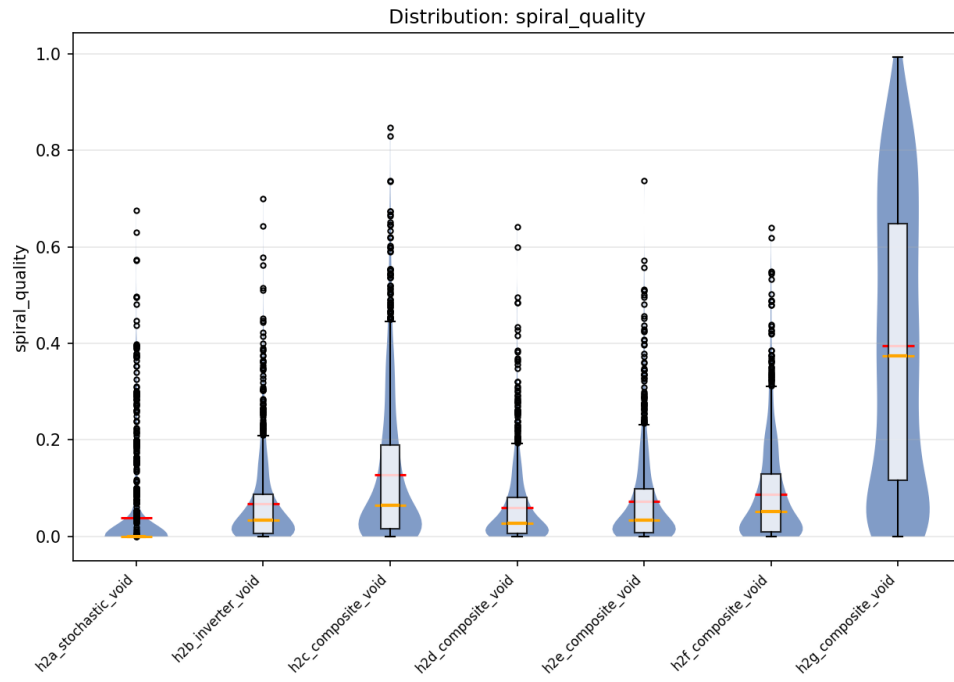


Figure S4. Distribution violin plots for H2: Specificity (Ablation). Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S5 / Figure S5: H3: Exploitation (Mode Switching)

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h3a_composite_dense_band]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0632	0.0322	0.0920	0.0779	3.32	0.0000	0.8778
food_eaten	1000	283.9310	324.0000	131.0305	143.2500	-0.94	0.0000	529.0000
area_cells_visited	1000	66.4640	68.0000	10.5267	11.0000	-0.36	25.0000	117.0000
mean_speed	1000	59.9622	55.9961	14.2879	13.1661	3.34	43.3279	204.5987

[h3b_composite_small_circle]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0921	0.0362	0.1423	0.0965	2.70	0.0000	0.9389
food_eaten	1000	144.5630	180.0000	107.1748	217.2500	-0.15	0.0000	357.0000
area_cells_visited	1000	69.1760	69.0000	14.1709	12.0000	0.84	20.0000	126.0000
mean_speed	1000	67.6897	61.9123	16.6626	26.2229	3.30	45.2702	260.3886

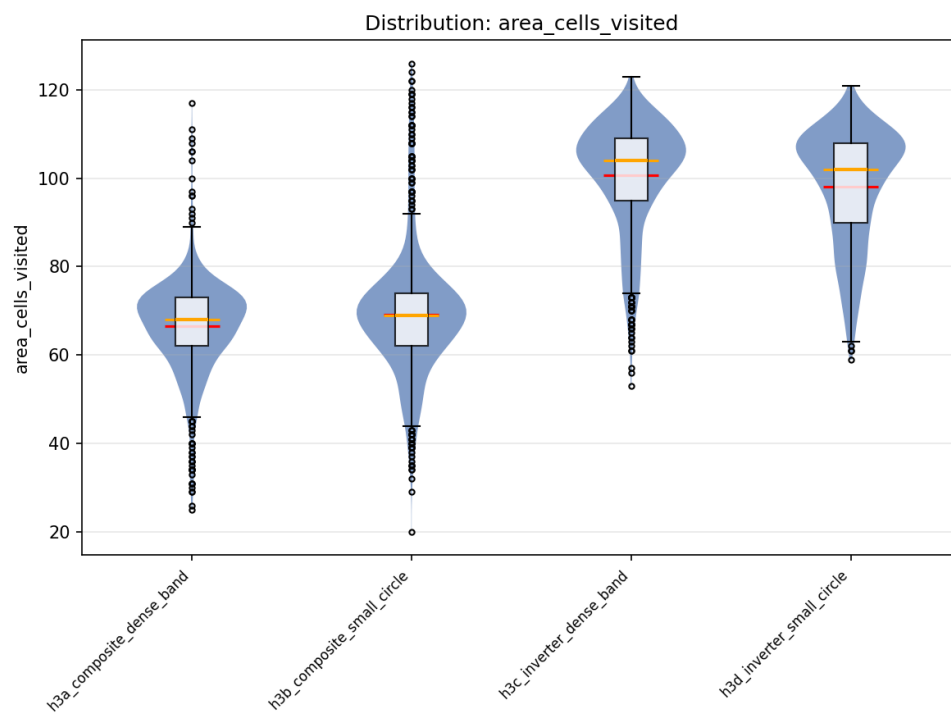
[h3c_inverter_dense_band]

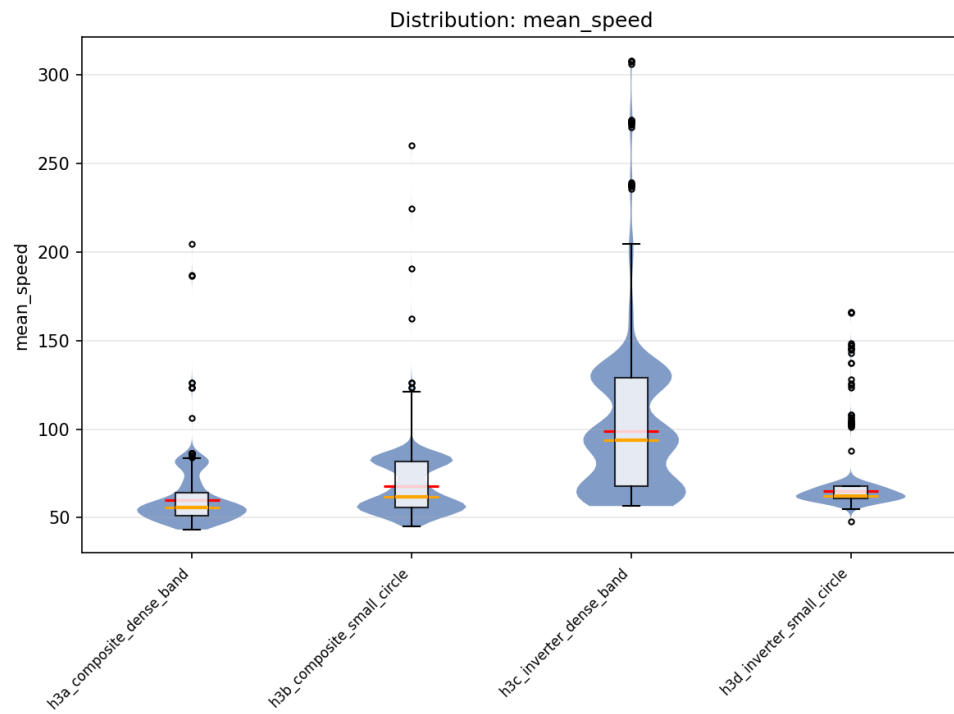
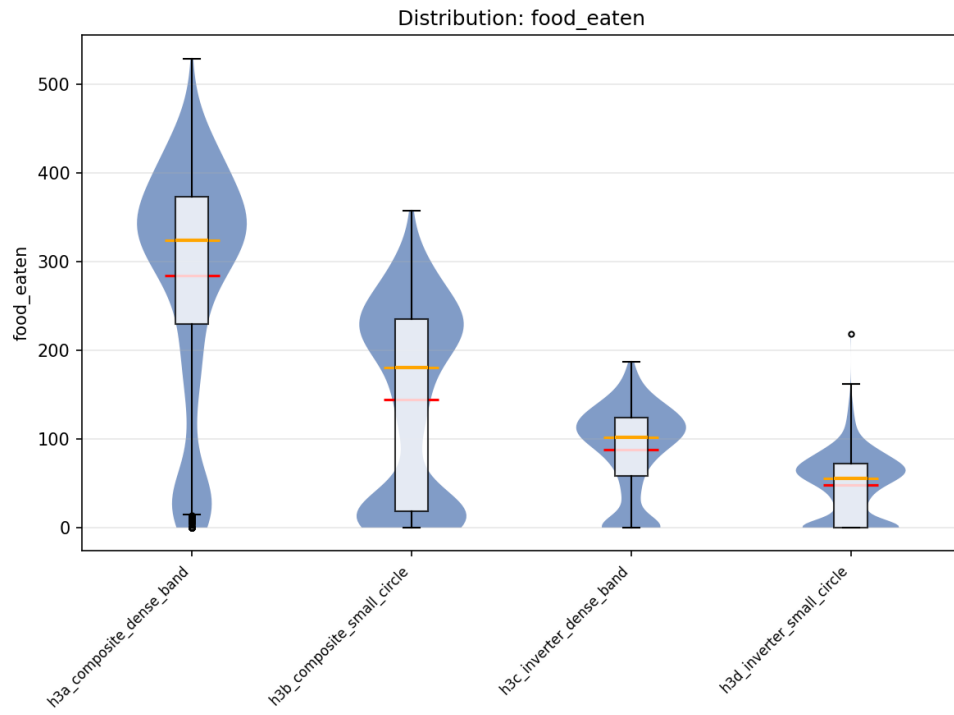
Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0708	0.0213	0.1138	0.0880	2.59	0.0000	0.9049
food_eaten	1000	88.1950	102.0000	50.3197	66.0000	-0.60	0.0000	187.0000
area_cells_visited	1000	100.7760	104.0000	12.3862	14.0000	-1.13	53.0000	123.0000
mean_speed	1000	99.0185	93.8467	39.3993	61.4466	1.94	56.6043	308.3184

[h3d_inverter_small_circle]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0650	0.0244	0.1068	0.0716	3.11	0.0000	0.7766
food_eaten	1000	48.2810	55.0000	35.6091	72.0000	0.13	0.0000	218.0000
area_cells_visited	1000	98.1620	102.0000	13.0463	18.0000	-0.84	59.0000	121.0000
mean_speed	1000	65.2544	62.1831	12.0258	6.6877	5.64	48.0818	166.3718

Table S5. Per-trial distribution statistics for H3: Exploitation (Mode Switching). $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.





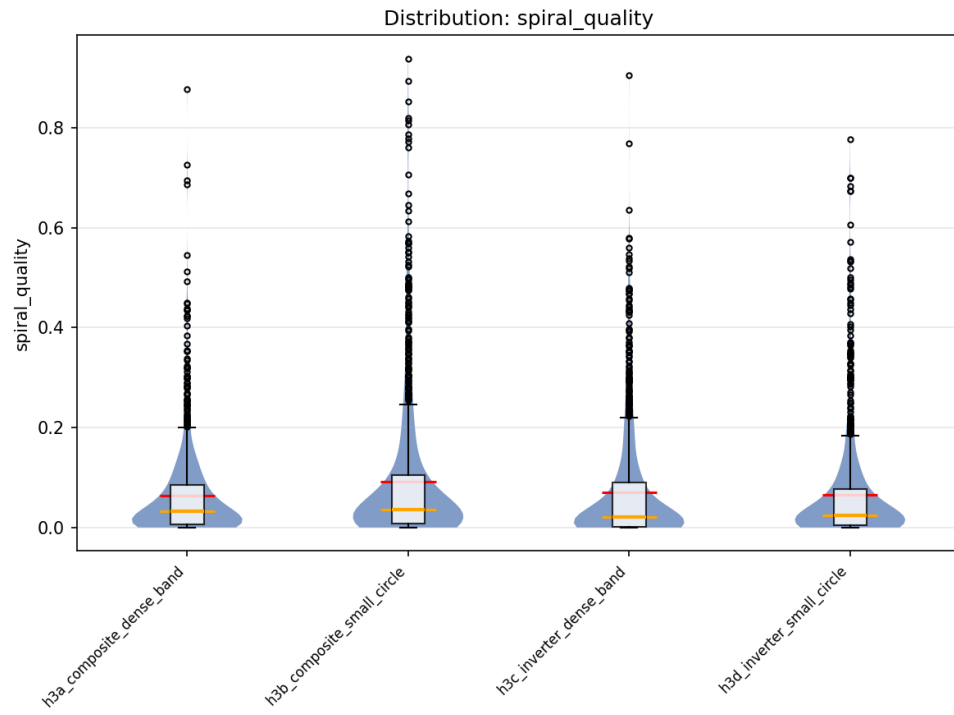


Figure S5. Distribution violin plots for H3: Exploitation (Mode Switching). Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S6 / Figure S6: H4a: Opposition Balance

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h4a1_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0709	0.0324	0.0951	0.0869	2.11	0.0000	0.6465
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	10.4120	10.0000	1.5915	2.0000	0.39	6.0000	16.0000
mean_speed	1000	97.9306	97.9306	0.0008	0.0011	0.20	97.9282	97.9334

[h4a2_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0884	0.0473	0.1068	0.1137	1.84	0.0000	0.6535
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	29.4890	29.0000	4.2042	6.0000	0.44	18.0000	45.0000
mean_speed	1000	102.6862	102.6862	0.0005	0.0007	-0.05	102.6845	102.6876

[h4a3_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1129	0.0659	0.1312	0.1446	1.79	0.0000	0.8514
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	74.7680	74.0000	9.7638	14.0000	0.06	46.0000	101.0000
mean_speed	1000	104.7791	104.7791	0.0003	0.0005	0.06	104.7780	104.7802

[h4a4_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2748	0.1979	0.2586	0.3976	0.83	0.0000	0.9643
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	118.0710	119.0000	7.5471	10.0000	-0.72	78.0000	137.0000
mean_speed	1000	123.1629	105.8747	35.3832	0.0008	2.14	105.8737	286.9531

[h4a5_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0720	0.0385	0.0965	0.0968	2.42	0.0000	0.6627
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	131.8670	132.0000	5.8861	7.0000	-0.78	104.0000	146.0000
mean_speed	1000	234.0953	219.5420	32.1679	30.2063	1.07	167.2062	458.8207

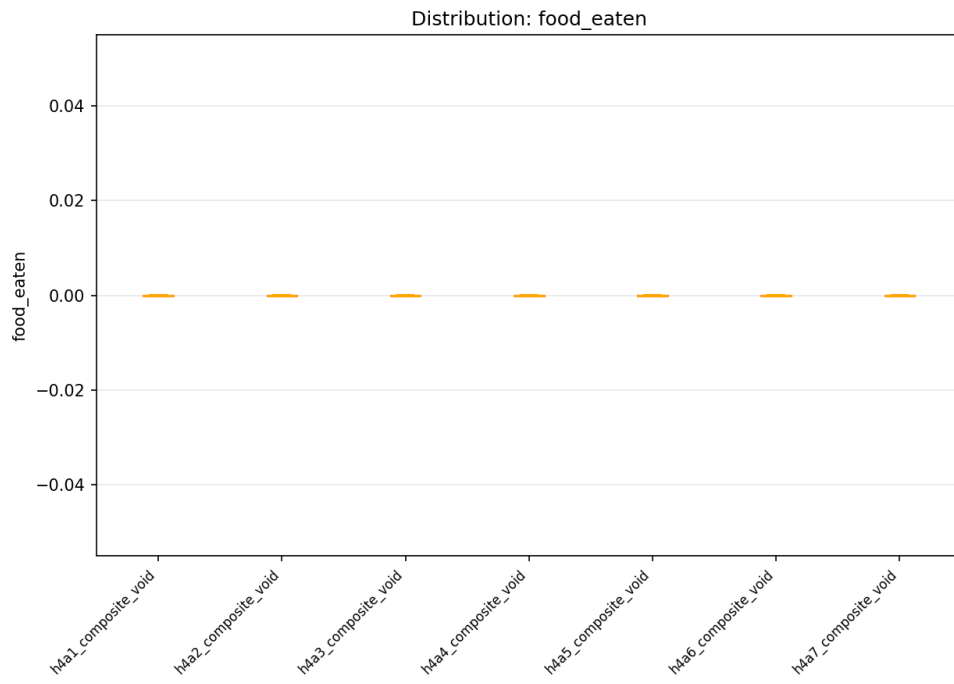
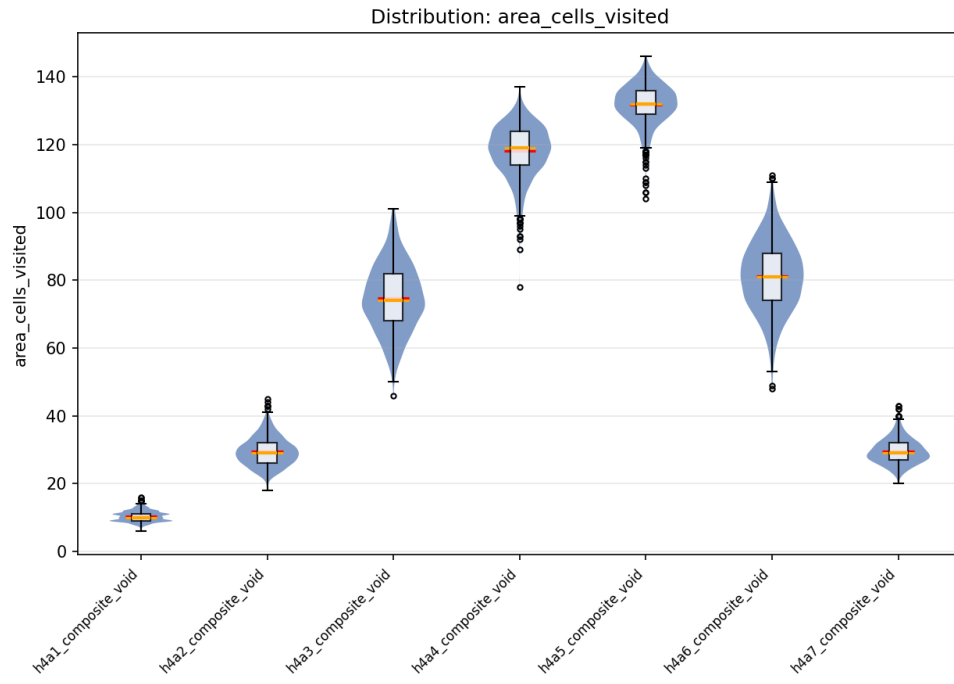
[h4a6_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1010	0.0604	0.1187	0.1270	1.85	0.0000	0.7254
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	81.2500	81.0000	10.2276	14.0000	-0.06	48.0000	111.0000
mean_speed	1000	109.6930	109.0594	6.6512	0.0005	11.78	109.0578	229.6900

[h4a7_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1139	0.0810	0.1092	0.1359	1.32	0.0000	0.6207
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	29.4560	29.0000	3.6223	5.0000	0.49	20.0000	43.0000
mean_speed	1000	112.2022	112.2022	0.0005	0.0007	-0.13	112.2003	112.2039

Table S6. Per-trial distribution statistics for H4a: Opposition Balance. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.



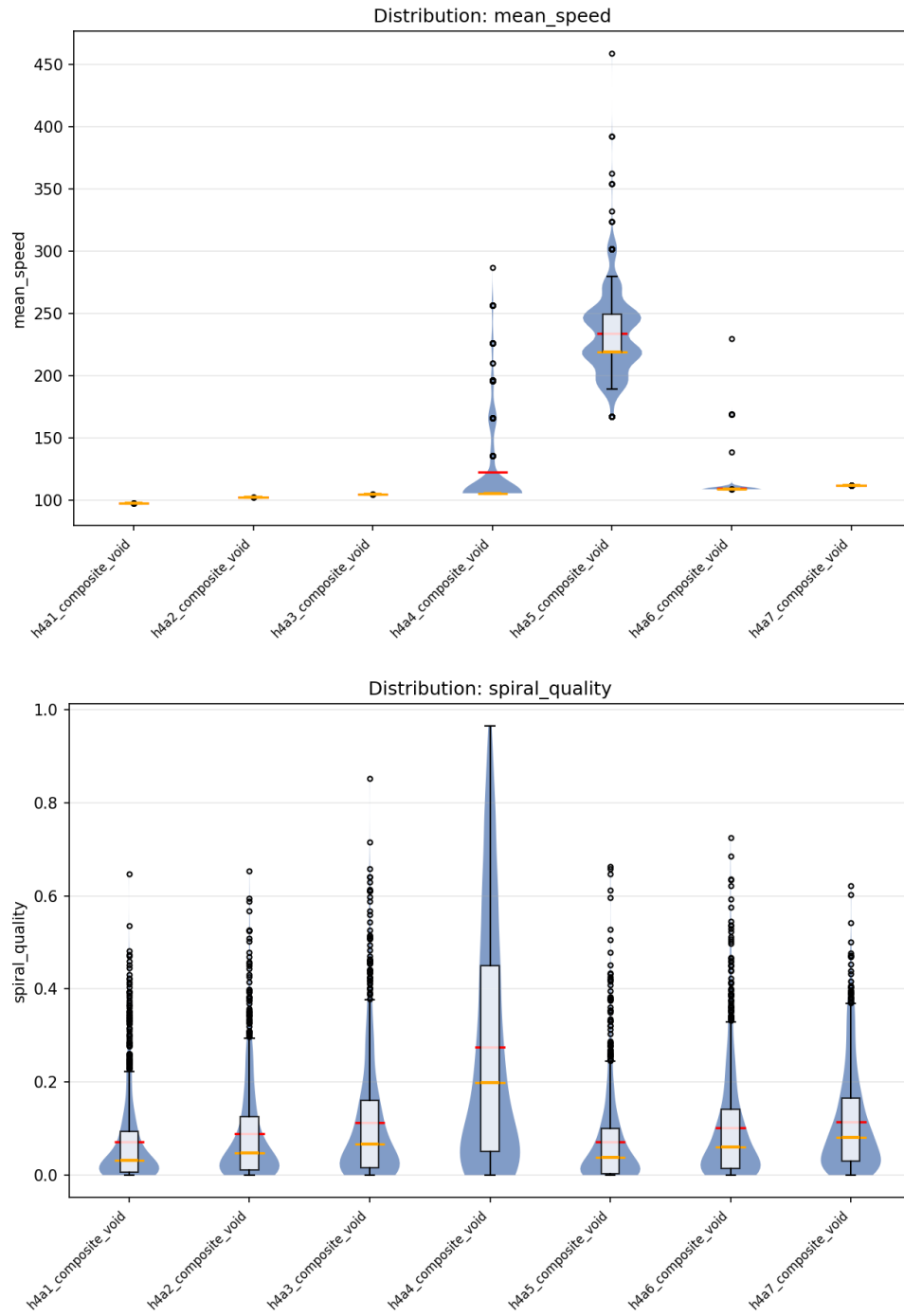


Figure S6. Distribution violin plots for H4a: Opposition Balance. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S7 / Figure S7: H4b: Robustness

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h4b1_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2369	0.1450	0.2411	0.3614	0.96	0.0000	0.9397
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	168.3410	170.0000	12.0653	16.0000	-0.45	124.0000	197.0000
mean_speed	1000	133.2684	106.5132	40.0365	40.2542	1.57	106.5122	366.3576

[h4b2_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1416	0.0699	0.1785	0.1956	1.71	0.0000	0.8855
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	125.4970	128.0000	57.6105	115.0000	-0.12	26.0000	212.0000
mean_speed	1000	143.9873	108.0078	58.8346	81.9374	1.27	102.7176	346.5854

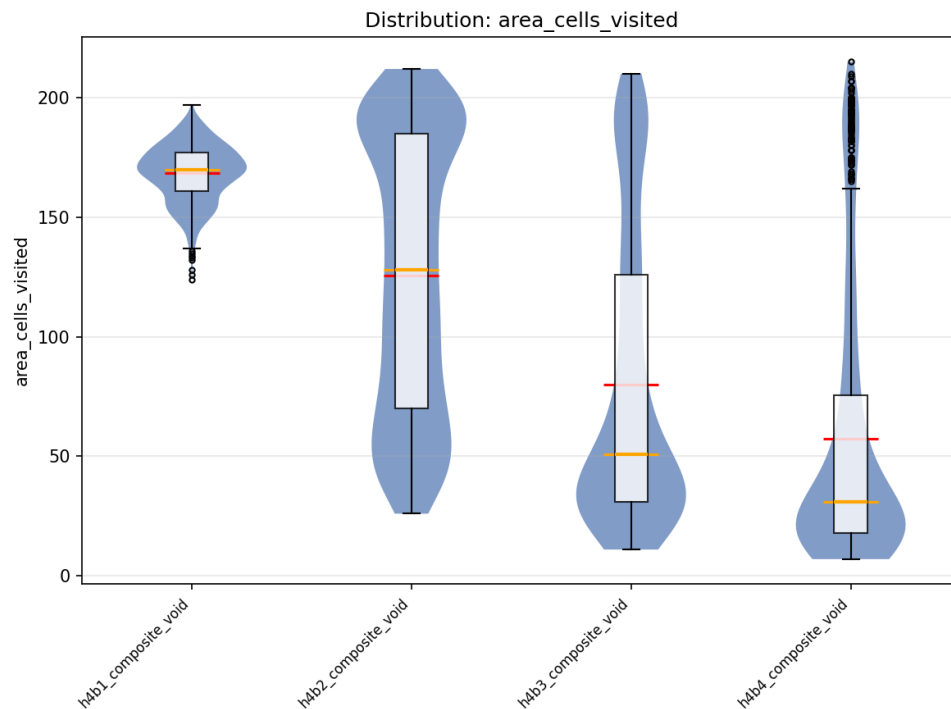
[h4b3_composite_void]

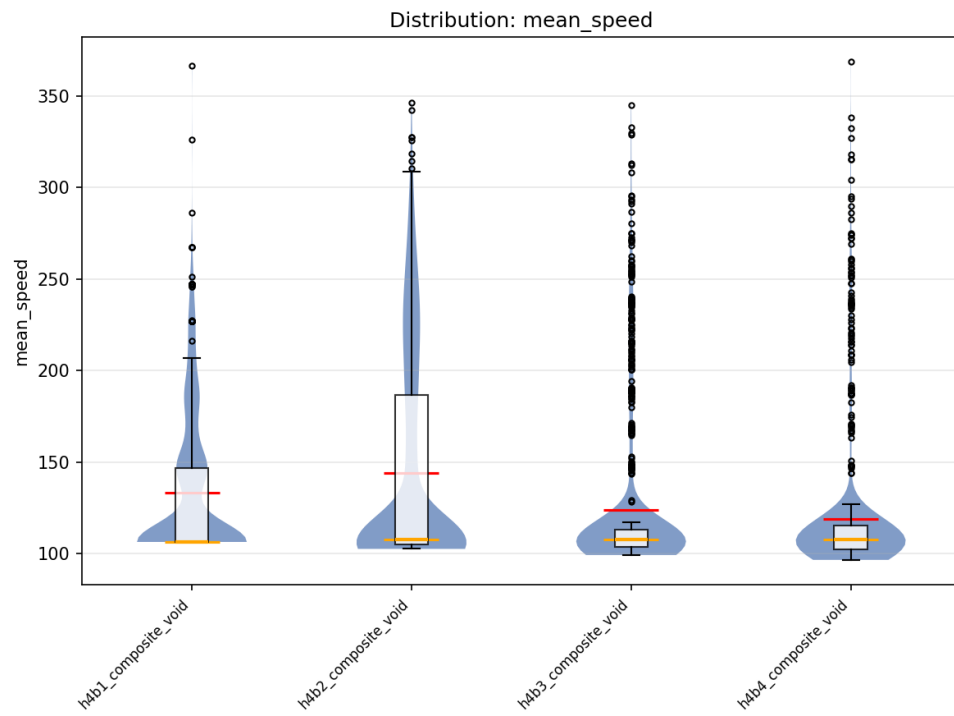
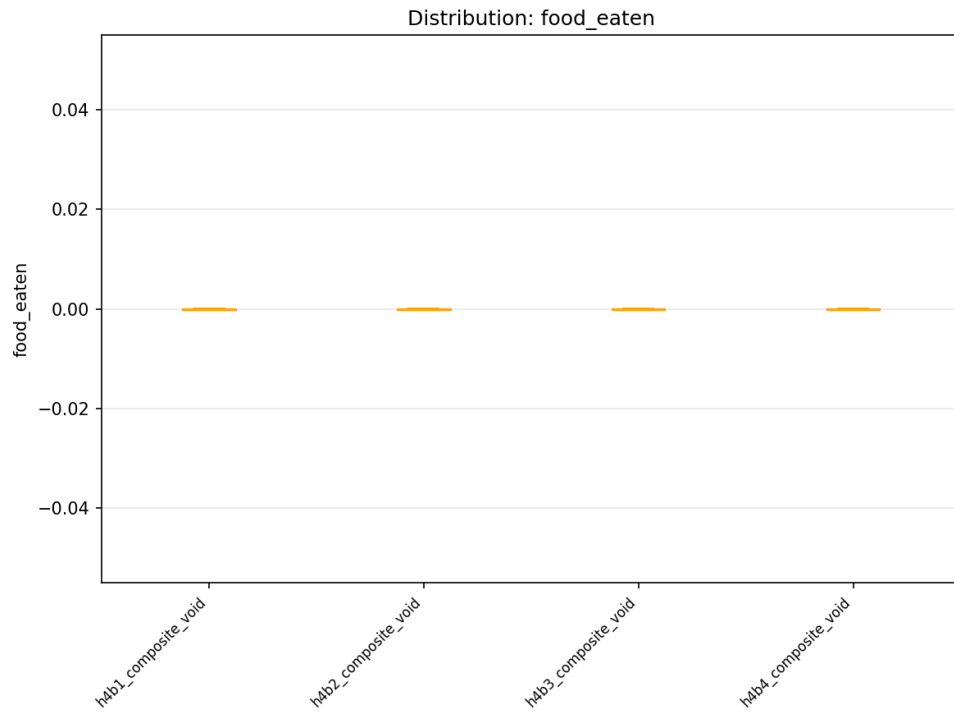
Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1049	0.0508	0.1388	0.1254	2.39	0.0000	0.9501
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	80.1080	51.0000	61.1591	95.0000	0.86	11.0000	210.0000
mean_speed	1000	123.8187	107.6237	43.6650	9.6622	2.58	99.3660	345.1814

[h4b4_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0914	0.0443	0.1209	0.1224	2.36	0.0000	0.8360
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	57.2710	31.0000	56.9988	57.7500	1.40	7.0000	215.0000
mean_speed	1000	118.9187	107.6774	38.5095	13.2014	3.43	96.7094	368.6344

Table S7. Per-trial distribution statistics for H4b: Robustness. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.





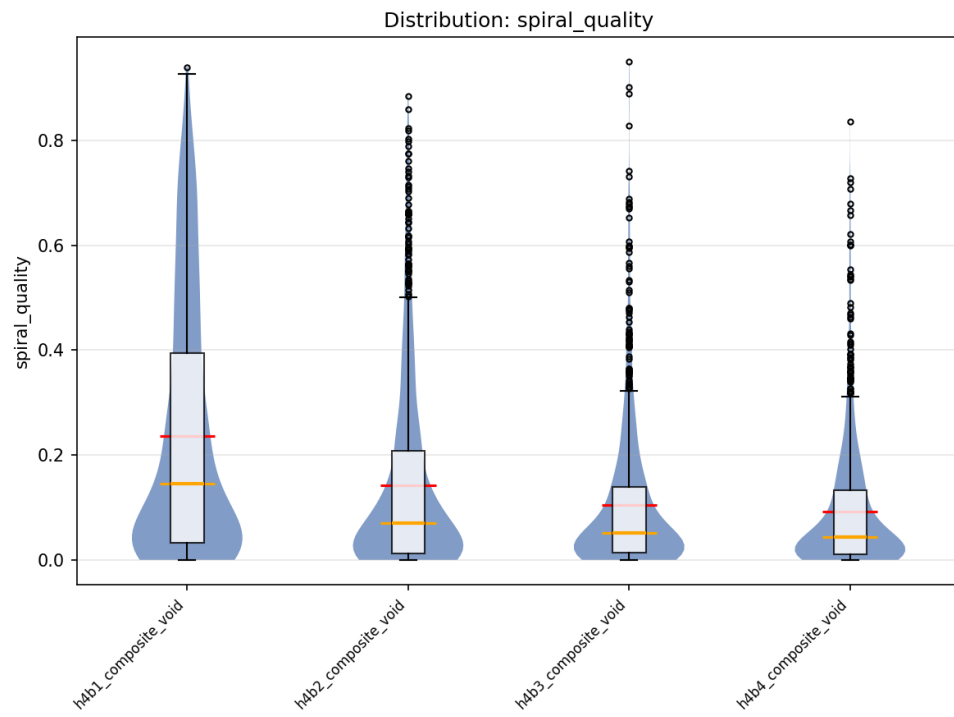


Figure S7. Distribution violin plots for H4b: Robustness. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S8 / Figure S8: H5: Exploitation Geometry

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h5a1_composite_dense_bands]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2221	0.2227	0.1186	0.1658	0.14	0.0000	0.6067
food_eaten	1000	456.2960	474.0000	127.0362	96.2500	-1.88	0.0000	767.0000
area_cells_visited	1000	48.9240	49.0000	10.9269	10.0000	-0.81	12.0000	83.0000
mean_speed	1000	156.9728	156.2587	12.2274	10.2903	1.23	122.2749	197.9292

[h5a2_composite_dense_bands]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1324	0.1153	0.1049	0.1700	0.65	0.0000	0.5087
food_eaten	1000	559.5830	552.0000	112.2796	144.0000	-0.18	136.0000	801.0000
area_cells_visited	1000	78.3450	78.0000	12.1089	17.0000	0.14	38.0000	121.0000
mean_speed	1000	150.3026	151.4622	13.7302	19.7038	-0.10	119.4549	197.4428

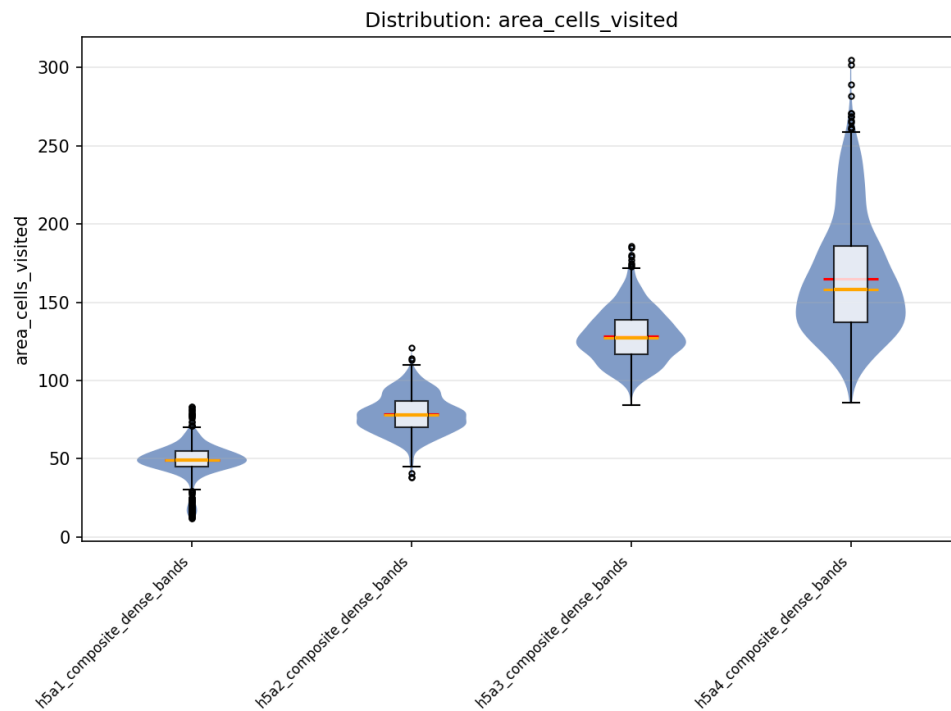
[h5a3_composite_dense_bands]

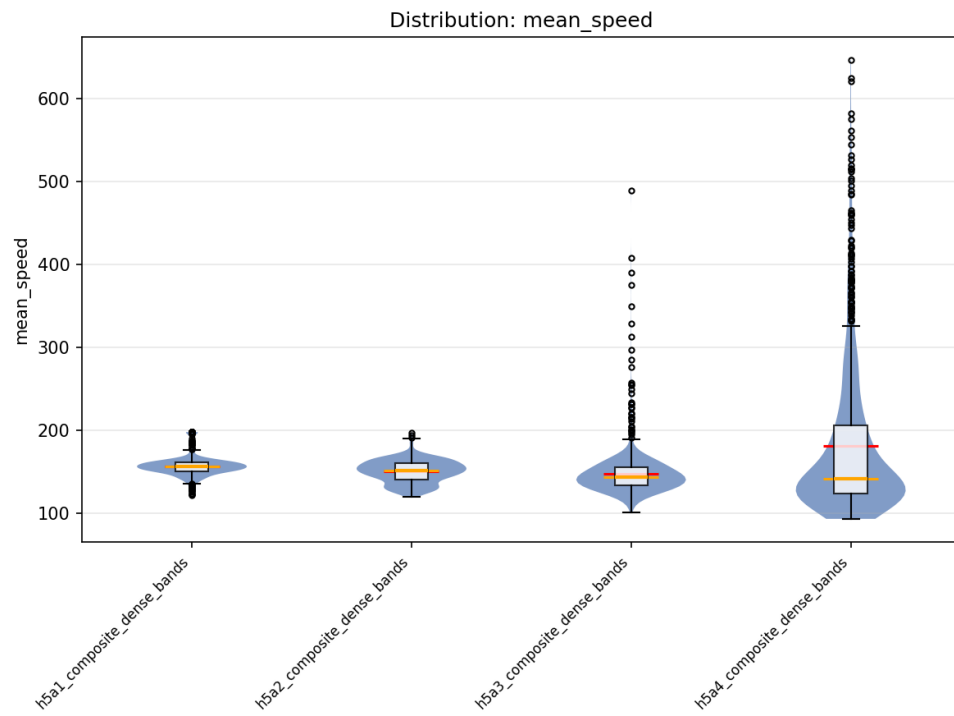
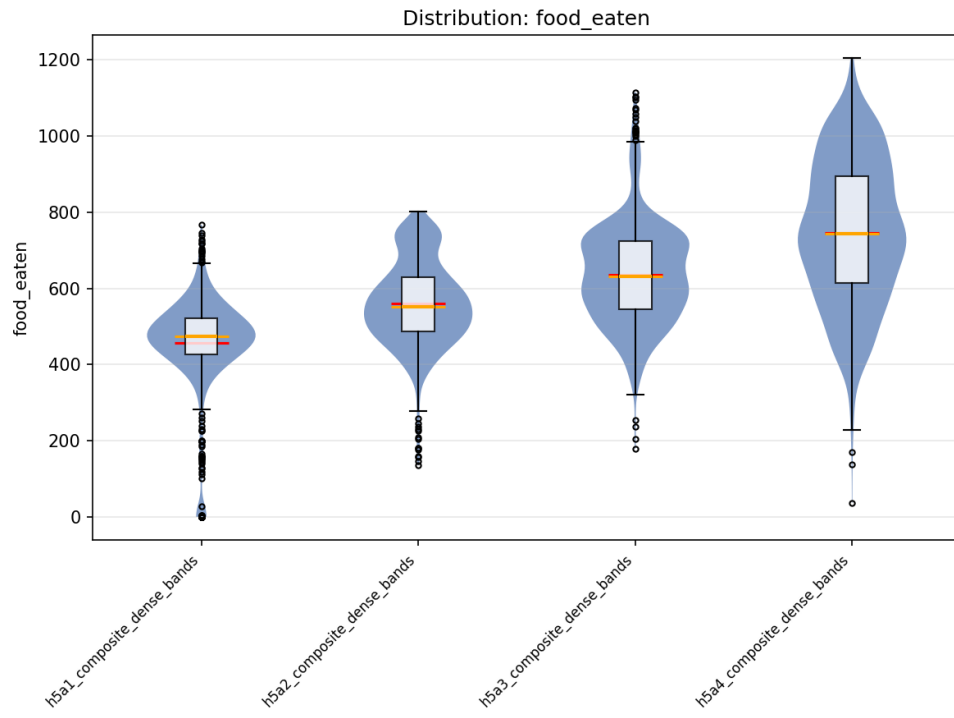
Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0583	0.0325	0.0693	0.0763	1.90	0.0000	0.5331
food_eaten	1000	636.5390	630.5000	133.9707	177.2500	0.44	179.0000	1114.0000
area_cells_visited	1000	128.3460	127.0000	16.4538	22.0000	0.40	84.0000	186.0000
mean_speed	1000	147.6598	143.3795	28.5019	22.4111	5.19	101.3059	488.5729

[h5a4_composite_dense_bands]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0750	0.0297	0.1092	0.0842	2.34	0.0000	0.6096
food_eaten	1000	746.5790	744.0000	190.8208	281.5000	-0.17	37.0000	1205.0000
area_cells_visited	1000	165.1760	158.0000	37.6160	49.0000	0.78	86.0000	305.0000
mean_speed	1000	181.2233	141.6257	91.4031	81.8273	2.09	93.3805	646.4613

Table S8. Per-trial distribution statistics for H5: Exploitation Geometry. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.





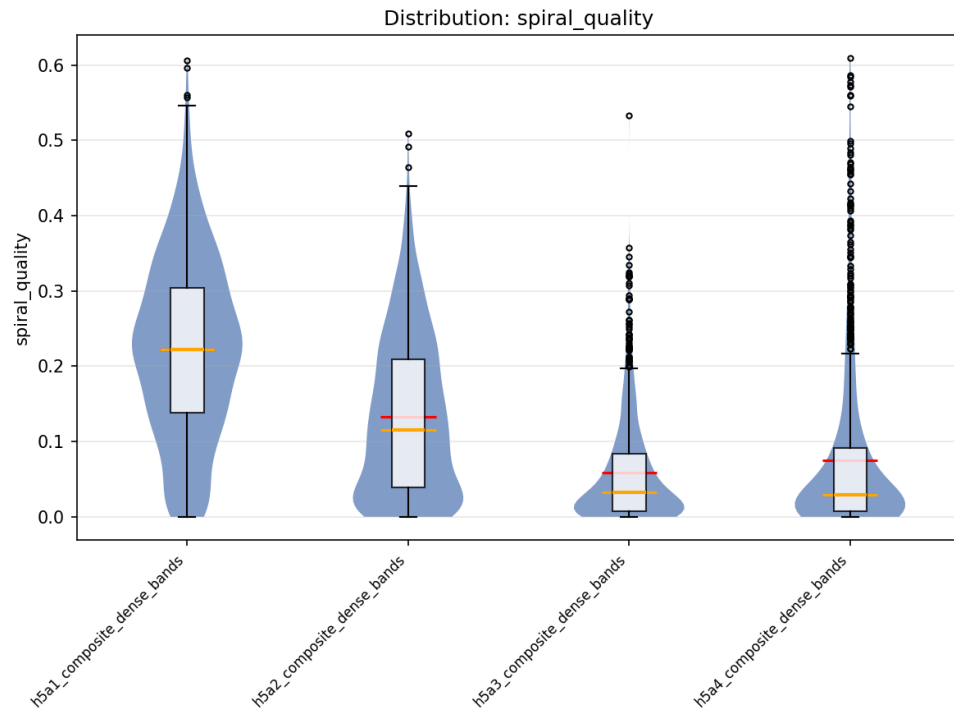


Figure S8. Distribution violin plots for H5: Exploitation Geometry. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S9 / Figure S9: H6: Complexity Scaling

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h6a1_composite_dense_bands]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0756	0.0357	0.1085	0.0856	2.62	0.0000	0.8161
food_eaten	1000	748.9880	736.0000	192.6669	293.2500	-0.05	189.0000	1226.0000
area_cells_visited	1000	165.1590	159.0000	36.3634	49.0000	0.67	89.0000	297.0000
mean_speed	1000	181.4854	142.3212	94.0227	85.1548	2.40	92.5846	744.2315

[h6a2_composite_dense_bands]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1243	0.0594	0.1570	0.1635	1.96	0.0000	0.8639
food_eaten	1000	733.8080	732.5000	179.3034	214.0000	-0.27	126.0000	1195.0000
area_cells_visited	1000	165.0790	156.0000	41.6126	49.0000	1.13	91.0000	313.0000
mean_speed	1000	209.2966	171.5859	95.6460	60.3007	2.93	128.0575	838.8190

[h6a3_composite_dense_bands]

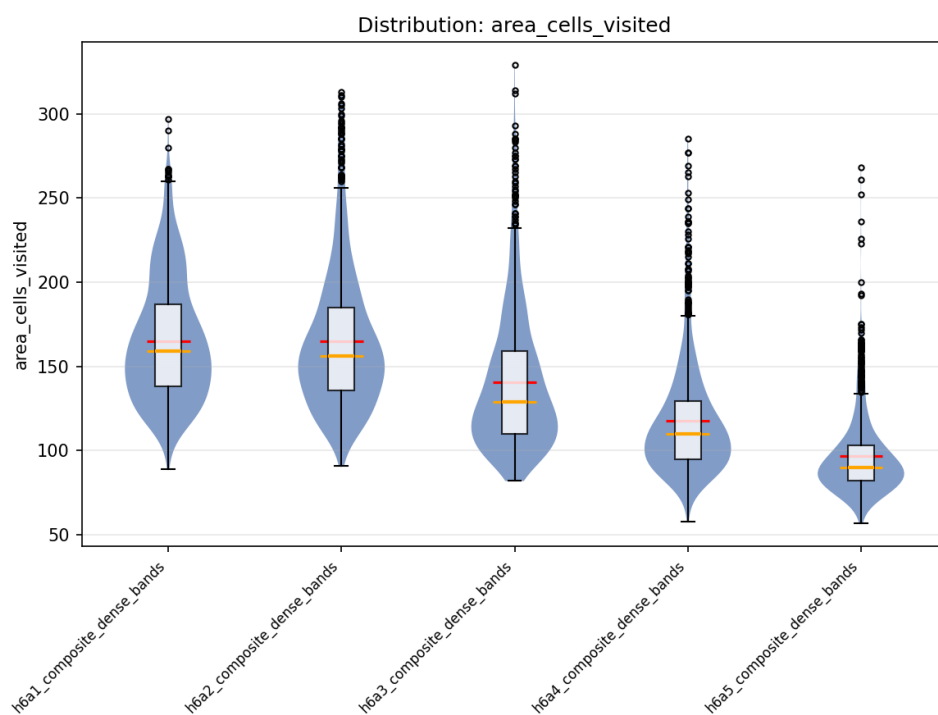
Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1509	0.0754	0.1735	0.2234	1.34	0.0000	0.8598
food_eaten	1000	654.7380	657.5000	140.9929	158.5000	-0.21	108.0000	1154.0000
area_cells_visited	1000	140.4780	129.0000	41.5432	49.0000	1.37	82.0000	329.0000
mean_speed	1000	208.3153	176.9853	81.6864	50.0265	3.32	136.5307	844.1364

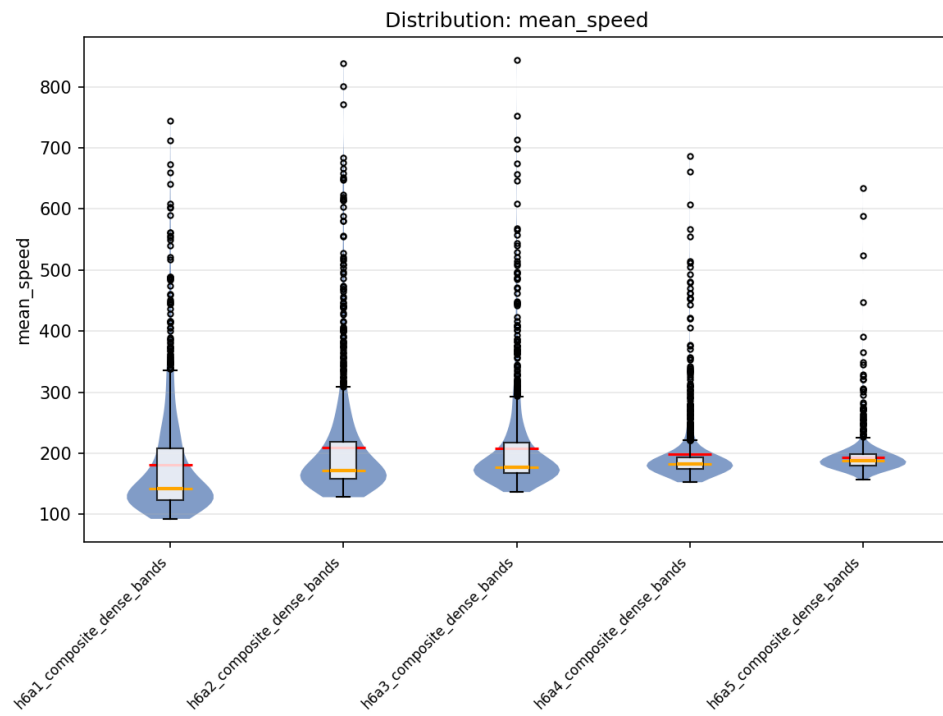
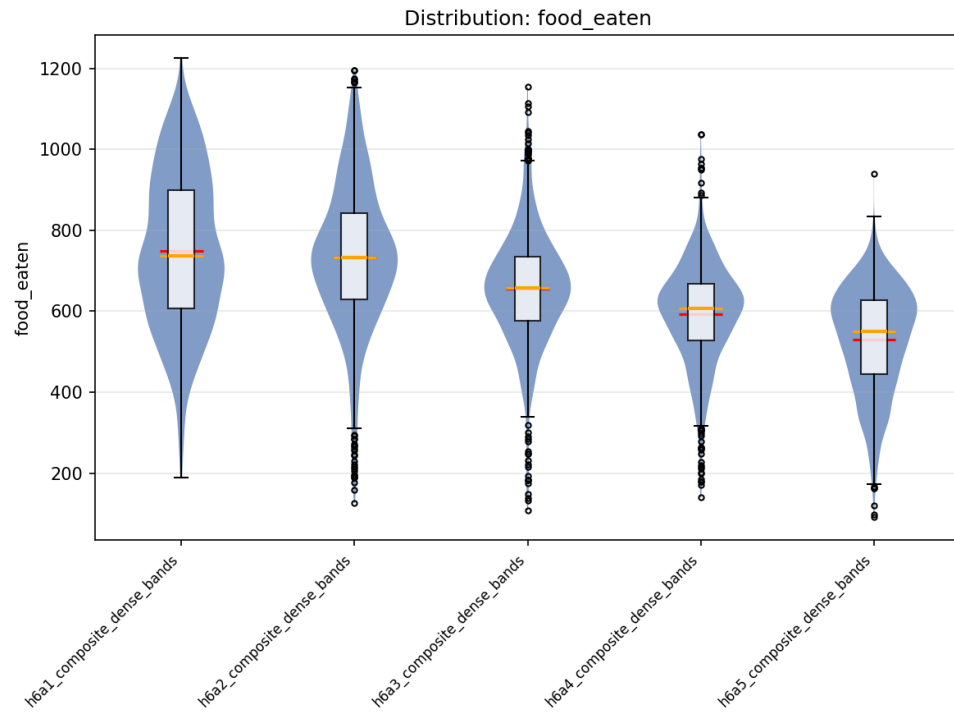
[h6a4_composite_dense_bands]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1607	0.0868	0.1800	0.2349	1.36	0.0000	0.9007
food_eaten	1000	593.1280	608.0000	124.7120	141.2500	-0.41	140.0000	1037.0000
area_cells_visited	1000	117.5750	110.0000	33.7300	34.2500	1.77	58.0000	285.0000
mean_speed	1000	197.9630	182.0885	56.2735	18.5151	4.36	152.4831	687.1341

[h6a5_composite_dense_bands]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1542	0.0816	0.1765	0.2202	1.40	0.0000	0.9415
food_eaten	1000	530.2490	550.5000	130.4715	182.2500	-0.55	92.0000	939.0000
area_cells_visited	1000	96.6480	90.0000	24.5981	21.0000	2.32	57.0000	268.0000
mean_speed	1000	193.4766	187.3396	32.8443	18.3982	6.92	157.4749	634.5070

Table S9. Per-trial distribution statistics for H6: Complexity Scaling. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.



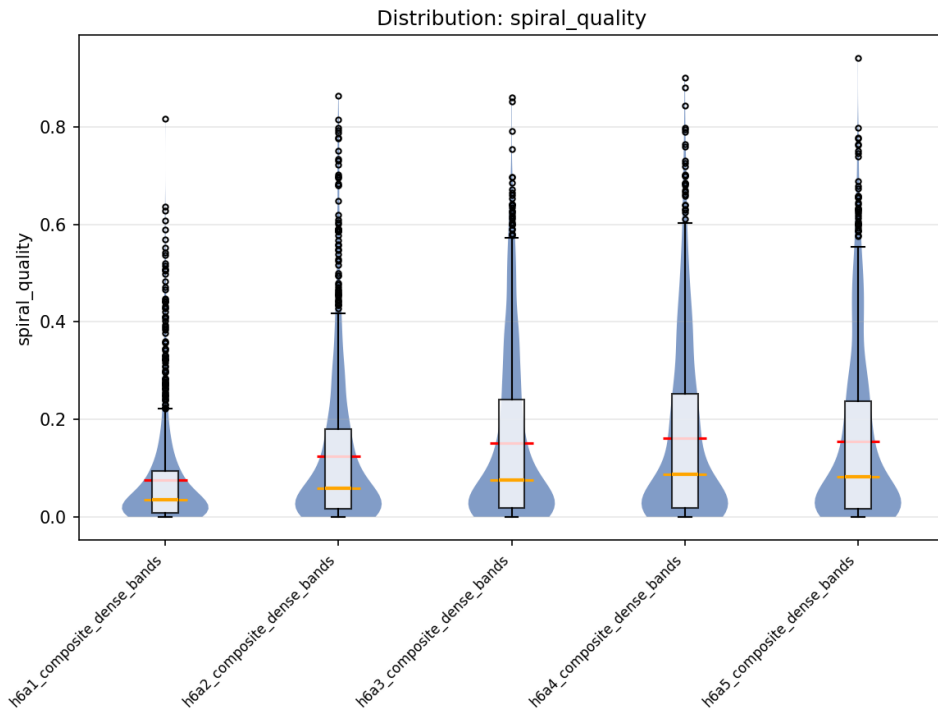


Figure S9. Distribution violin plots for H6: Complexity Scaling. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S10 / Figure S10: H7: Noise Sensitivity

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h7a1_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.6521	0.6991	0.1345	0.0016	-2.74	0.1738	0.7109
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	63.3270	63.0000	1.9769	3.0000	0.17	58.0000	70.0000
mean_speed	1000	153.7312	153.7312	0.0003	0.0004	0.08	153.7303	153.7321

[h7a2_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.3662	0.3519	0.2404	0.3903	0.28	0.0000	0.9529
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	101.2790	101.5000	10.1094	14.0000	-0.20	62.0000	128.0000
mean_speed	1000	153.8921	153.7312	3.5951	0.0004	22.33	153.7301	234.2507

[h7a3_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2888	0.2350	0.2526	0.3885	0.75	0.0000	0.9816
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	113.8430	115.0000	9.0674	13.0000	-0.49	73.0000	138.0000
mean_speed	1000	159.2741	153.7312	22.2624	0.0006	4.18	153.7302	314.3148

[h7a4_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.3058	0.2396	0.2672	0.4645	0.64	0.0000	0.9846
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	117.8780	118.0000	8.5637	12.0000	-0.30	89.0000	139.0000
mean_speed	1000	170.1429	153.7313	39.7030	0.0008	2.60	153.7299	394.6841

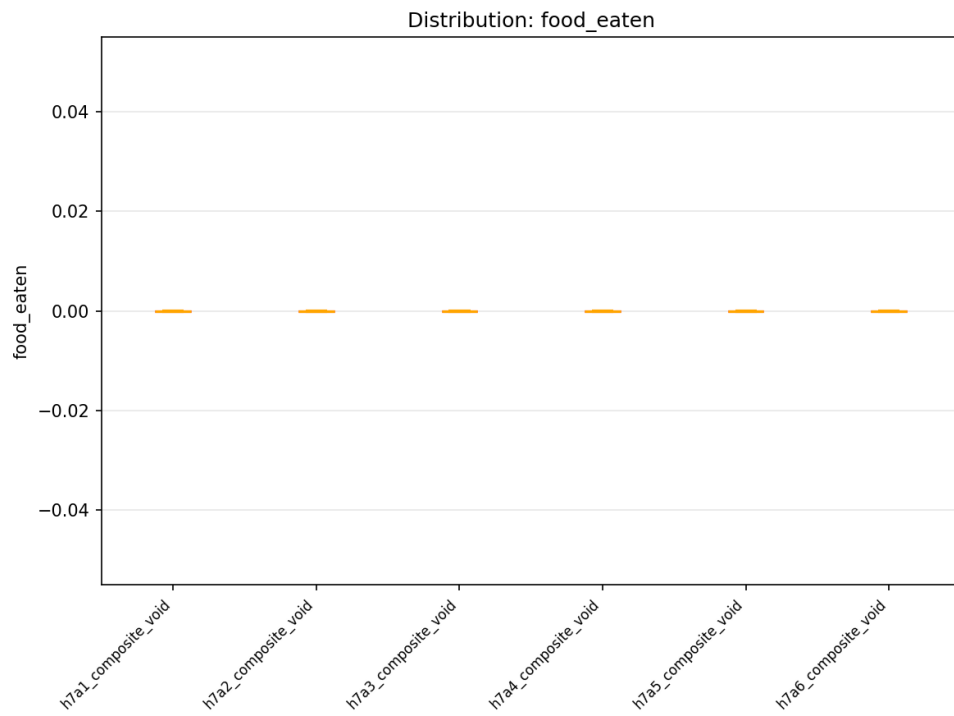
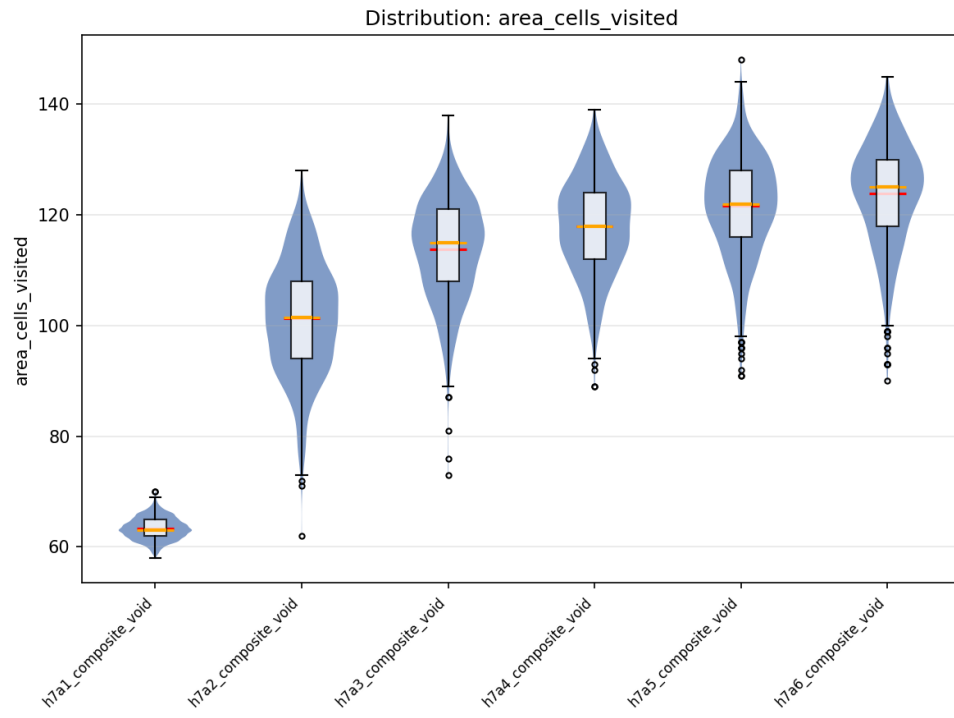
[h7a5_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.3078	0.2551	0.2621	0.4488	0.53	0.0000	0.9823
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	121.5980	122.0000	9.0282	12.0000	-0.45	91.0000	148.0000
mean_speed	1000	190.2710	153.7316	57.2503	80.2617	1.56	153.7295	482.8630

[h7a6_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2380	0.1889	0.2118	0.3296	0.79	0.0000	0.8950
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	123.8040	125.0000	9.1649	12.0000	-0.59	90.0000	145.0000
mean_speed	1000	216.2842	193.8317	79.7520	120.3193	1.53	153.7289	691.5763

Table S10. Per-trial distribution statistics for H7: Noise Sensitivity. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.



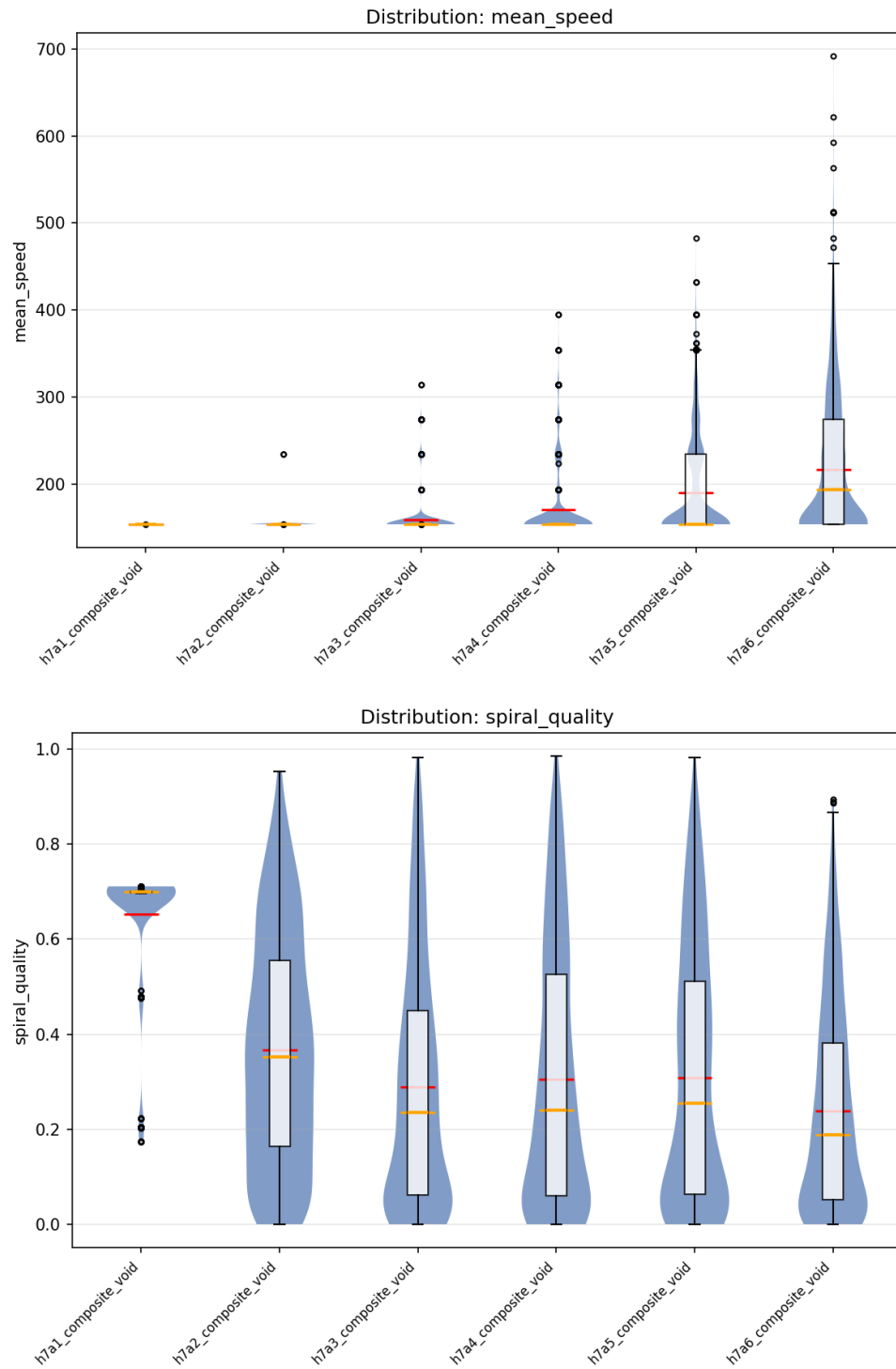


Figure S10. Distribution violin plots for H7: Noise Sensitivity. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S11 / Figure S11: H8: Reconciliation

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[h8a1_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.2703	0.1975	0.2503	0.3791	0.84	0.0000	0.9674
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	164.5180	166.0000	11.1187	14.0000	-0.55	126.0000	192.0000
mean_speed	1000	268.0486	213.0583	80.7992	80.2279	1.39	213.0565	573.0999

[h8a2_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.3101	0.2388	0.2686	0.4671	0.54	0.0000	0.9395
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	190.5410	192.0000	12.2702	16.0000	-0.61	135.0000	218.0000
mean_speed	1000	325.0267	322.1992	99.2878	159.7442	1.23	242.3913	898.9109

[h8a3_composite_void]

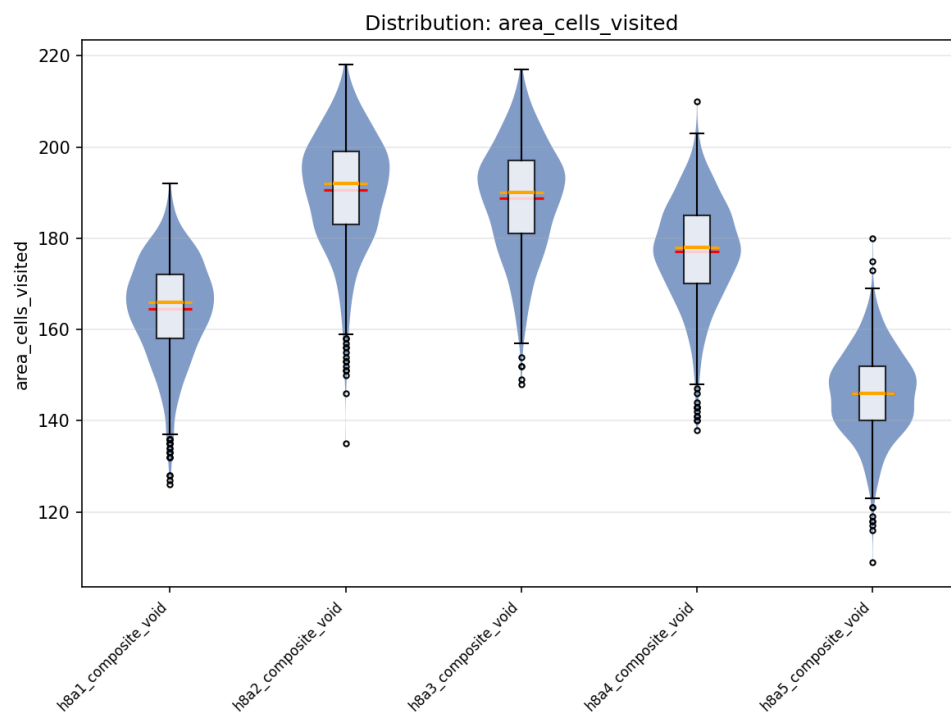
Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.3826	0.3696	0.2751	0.4930	0.22	0.0000	0.9778
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	188.6930	190.0000	12.0042	16.0000	-0.41	148.0000	217.0000
mean_speed	1000	326.1351	326.0792	90.9432	138.8757	1.04	246.2565	763.1757

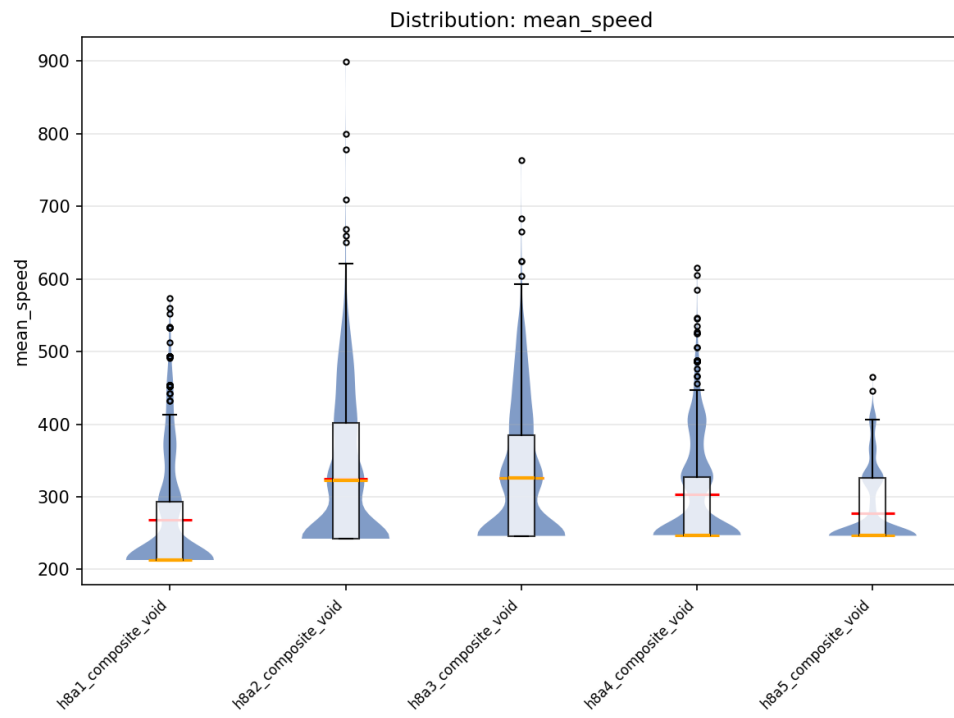
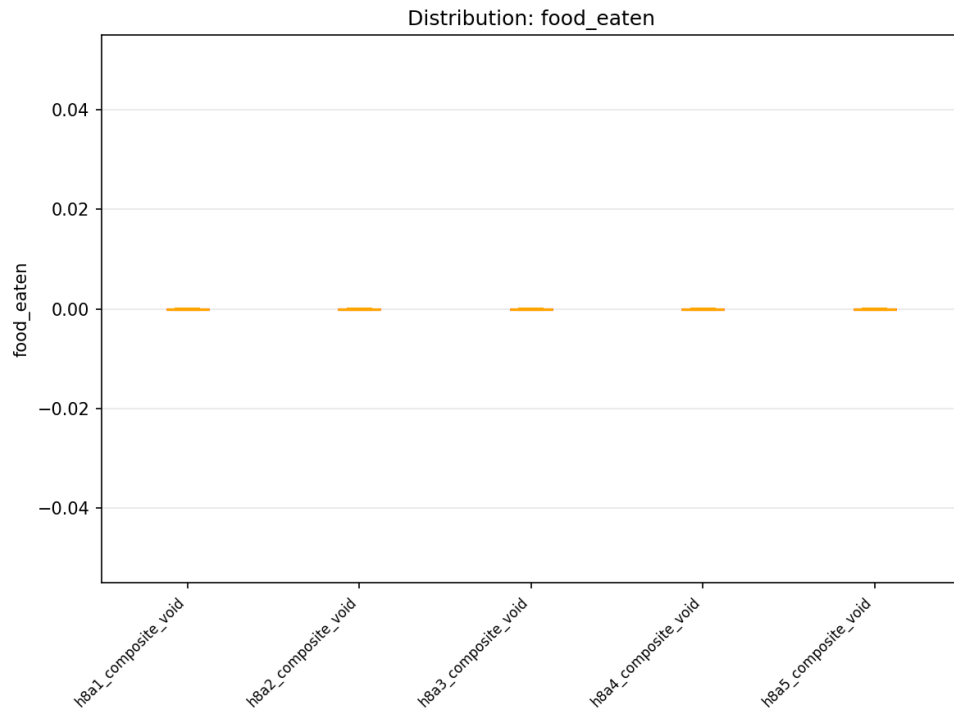
[h8a4_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.5383	0.5674	0.2639	0.4144	-0.27	0.0002	0.9935
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	177.1270	178.0000	11.0328	15.0000	-0.43	138.0000	210.0000
mean_speed	1000	303.2637	247.2714	72.8300	80.2095	1.16	247.2689	615.6298

[h8a5_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.7196	0.7545	0.1707	0.2272	-0.92	0.0822	0.9821
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	145.9640	146.0000	9.2290	12.0000	-0.14	109.0000	180.0000
mean_speed	1000	277.6242	246.4221	49.8443	79.8469	1.35	246.4199	464.9095

Table S11. Per-trial distribution statistics for H8: Reconciliation. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.



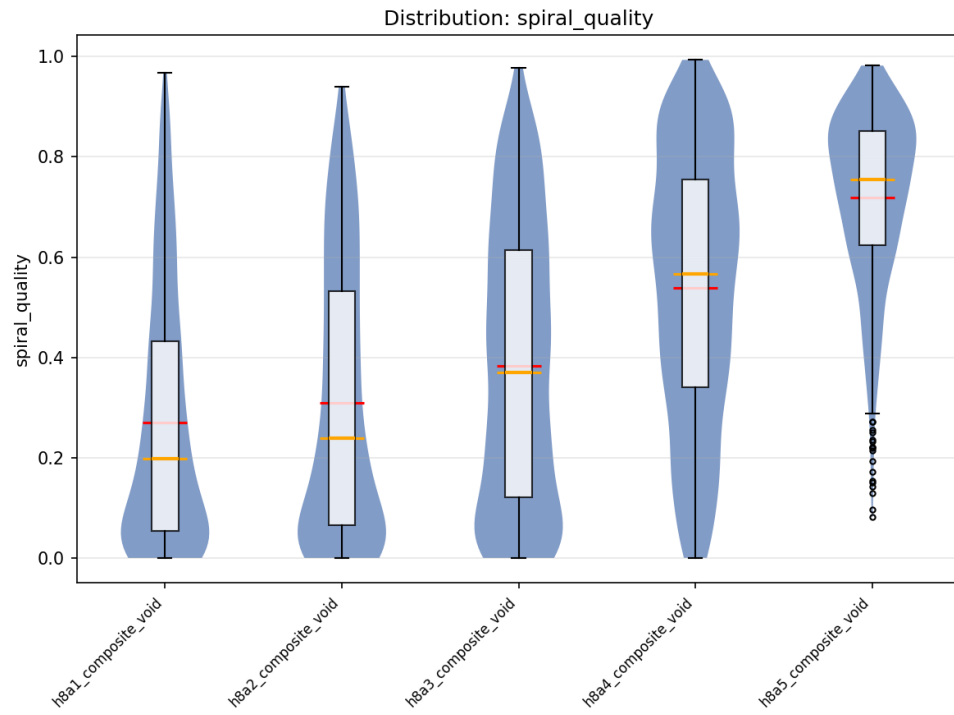


Figure S11. Distribution violin plots for H8: Reconciliation. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S12 / Figure S12: H9: Negative Controls

DISTRIBUTION REPORT - Per-Trial Metric Statistics

[neg_h0_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0252	0.0017	0.0483	0.0257	2.99	0.0000	0.4050
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	124.4470	129.0000	15.5580	19.0000	-2.15	31.0000	144.0000
mean_speed	1000	303.3359	312.3174	31.8923	50.9947	-0.25	253.3304	352.5449

[neg_h1_stochastic_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0654	0.0000	0.1035	0.0979	1.83	0.0000	0.5376
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	57.5510	58.0000	6.8736	9.0000	0.04	36.0000	77.0000
mean_speed	1000	122.1490	106.5025	43.9829	42.8836	1.21	41.6256	313.1645

[neg_h2_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1069	0.0453	0.1433	0.1349	2.08	0.0000	0.7606
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	147.1260	146.0000	21.7788	29.0000	0.20	87.0000	214.0000
mean_speed	1000	433.1839	420.8133	65.0636	0.0010	6.67	420.8113	1133.3890

[neg_h3_stochastic_dense_band]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0635	0.0000	0.1055	0.0979	1.92	0.0000	0.5981
food_eaten	1000	52.8890	39.0000	52.1503	56.2500	1.73	0.0000	346.0000
area_cells_visited	1000	57.6580	57.0000	6.6859	9.0000	0.20	36.0000	85.0000
mean_speed	1000	121.5133	106.5961	42.4073	41.2549	1.31	46.5080	394.0852

[neg_h4a_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0961	0.0507	0.1148	0.1300	1.61	0.0000	0.5764
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	5.8650	6.0000	1.1018	2.0000	0.20	4.0000	10.0000
mean_speed	1000	213.1441	213.1440	0.0026	0.0036	-0.03	213.1351	213.1516

[neg_h4b_stochastic_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0556	0.0000	0.1008	0.0855	2.14	0.0000	0.5349
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	56.8970	57.0000	6.7648	9.0000	0.00	33.0000	77.0000
mean_speed	1000	122.0322	106.5021	44.7202	42.9662	1.39	36.9303	382.2723

[neg_h5_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.1434	0.0731	0.1729	0.1971	1.78	0.0000	0.8543
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	94.0930	94.0000	13.1836	18.0000	0.23	57.0000	143.0000
mean_speed	1000	224.4069	210.9146	56.3117	0.0008	5.02	210.9129	690.9990

[neg_h6_composite_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.3716	0.3679	0.2753	0.4930	0.21	0.0000	0.9653
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	191.3860	192.0000	11.4401	14.0000	-0.49	143.0000	225.0000
mean_speed	1000	393.2039	385.0810	135.9134	239.6424	0.94	246.2564	1080.9797

[neg_h7_inverter_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0036	0.0033	0.0031	0.0035	1.64	0.0000	0.0222
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	10.4600	11.0000	1.0842	2.0000	-0.08	8.0000	12.0000
mean_speed	1000	140.2713	140.2713	0.0006	0.0008	-0.01	140.2689	140.2737

[neg_h8a_stochastic_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0752	0.0262	0.1019	0.1060	1.62	0.0000	0.5798
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000
area_cells_visited	1000	56.2940	56.0000	5.1287	7.0000	0.00	38.0000	74.0000
mean_speed	1000	109.0962	101.1465	45.9283	68.9700	1.17	46.1820	360.3090

[neg_h8b_stochastic_void]

Metric	N	Mean	Median	Std	IQR	Skew	Min	Max
spiral_quality	1000	0.0390	0.0000	0.0884	0.0039	2.75	0.0000	0.5584
food_eaten	1000	0.0000	0.0000	0.0000	0.0000	0.00	0.0000	0.0000

area_cells_visited	1000	56.5620	56.0000	9.1754	11.2500	0.20	25.0000	89.0000
mean_speed	1000	127.8448	118.2839	41.9159	67.1316	0.89	24.7621	342.4037

Table S12. Per-trial distribution statistics for H9: Negative Controls. $N = 1,000$ independent trials per configuration. Skew = Fisher's skewness coefficient.

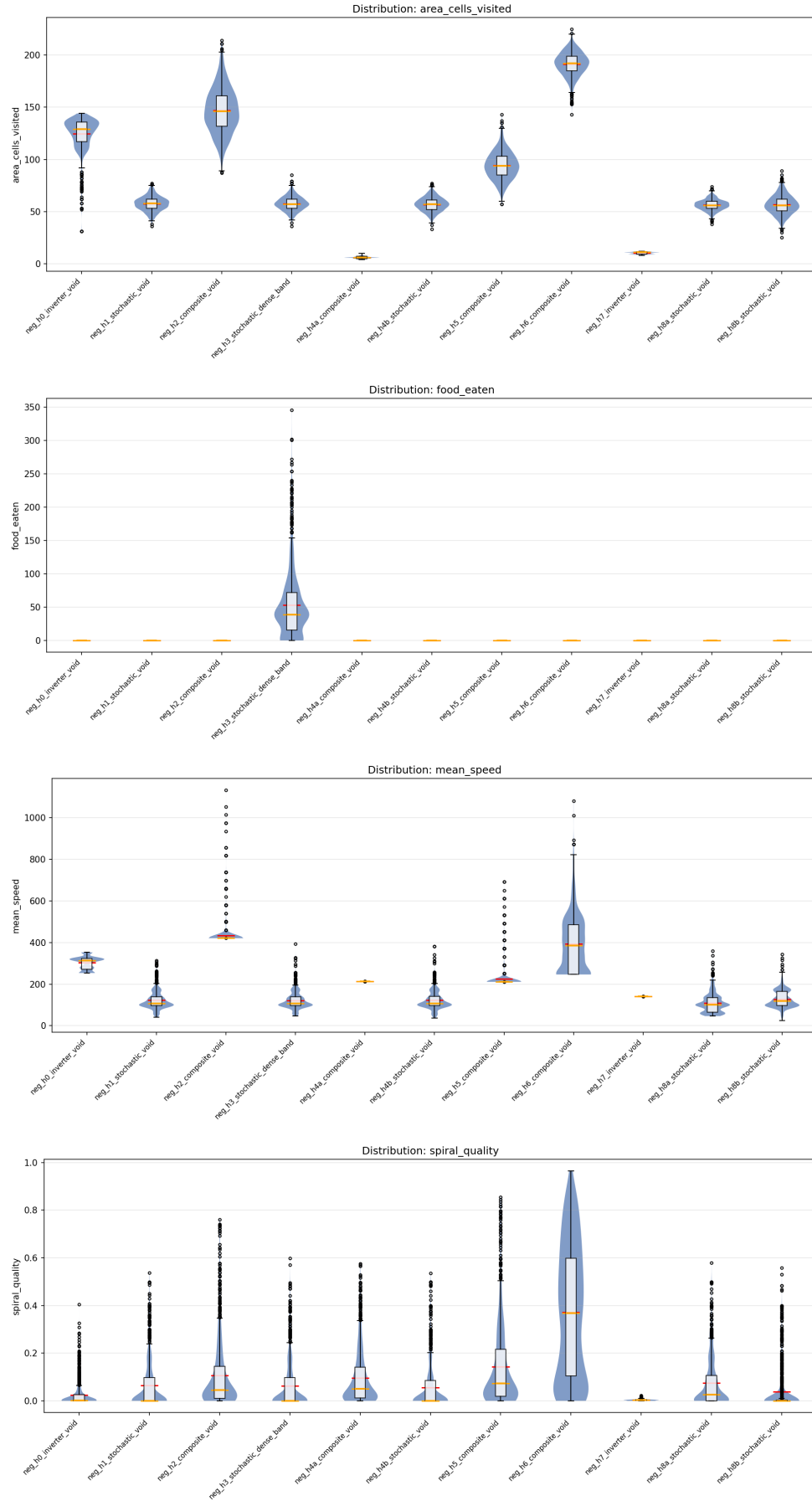


Figure S12. Distribution violin plots for H9: Negative Controls. Violin shows kernel density estimate; box shows IQR; red line = mean, orange line = median.

Table S13: External Baseline Comparison

Cohen's d effect sizes for NNN 3-inverter and 7-inverter composites versus five standard foraging baselines (stochastic random walk, correlated random walk, Lévy walk, prescribed logarithmic spiral, area-restricted search). Each agent was tested for 1,000 trials per environment. Effect size magnitudes: negligible (<0.2), small (0.2–0.5), medium (0.5–0.8), large (0.8–1.2), very large (>1.2).

BENCHMARK BASELINE COMPARISON

Environment: void

Agent	food_eaten	area_visited	spiral_quality	mean_speed
nnn3	0.00 +/- 0.00	166.91 +/- 12.18	0.25 +/- 0.24	133.29 +/- 40.58
nnn7	0.00 +/- 0.00	149.57 +/- 9.18	0.67 +/- 0.20	138.39 +/- 24.55
stochastic	0.00 +/- 0.00	57.32 +/- 6.94	0.08 +/- 0.11	61.16 +/- 22.75
crw	0.00 +/- 0.00	113.67 +/- 4.28	0.07 +/- 0.09	127.71 +/- 35.69
levy	0.00 +/- 0.00	110.36 +/- 6.77	0.07 +/- 0.09	123.72 +/- 44.58
logspiral	0.00 +/- 0.00	131.06 +/- 2.78	0.33 +/- 0.42	88.84 +/- 16.07
ars	0.00 +/- 0.00	113.10 +/- 4.22	0.07 +/- 0.09	130.54 +/- 28.70

Environment: dense_band

Agent	food_eaten	area_visited	spiral_quality	mean_speed
nnn3	372.53 +/- 250.10	114.48 +/- 37.35	0.12 +/- 0.14	185.80 +/- 140.20
nnn7	213.12 +/- 141.63	108.06 +/- 28.51	0.19 +/- 0.19	274.44 +/- 136.16
stochastic	44.37 +/- 49.87	57.37 +/- 6.72	0.07 +/- 0.10	73.13 +/- 31.43
crw	87.69 +/- 70.54	114.08 +/- 4.02	0.06 +/- 0.08	145.03 +/- 45.83
levy	87.28 +/- 76.51	111.29 +/- 6.67	0.06 +/- 0.08	155.72 +/- 60.38
logspiral	105.84 +/- 46.07	133.09 +/- 2.49	0.60 +/- 0.16	204.19 +/- 24.06
ars	119.59 +/- 53.16	83.12 +/- 16.99	0.19 +/- 0.17	110.55 +/- 42.70

Environment: dense_bands

Agent	food_eaten	area_visited	spiral_quality	mean_speed
nnn3	482.85 +/- 76.36	71.02 +/- 15.96	0.07 +/- 0.08	62.03 +/- 20.34
nnn7	385.25 +/- 131.48	50.51 +/- 8.30	0.13 +/- 0.16	82.71 +/- 10.79
stochastic	61.19 +/- 61.28	57.45 +/- 6.95	0.07 +/- 0.10	63.13 +/- 25.46
crw	105.11 +/- 79.38	113.62 +/- 4.03	0.06 +/- 0.08	130.87 +/- 37.71
levy	126.89 +/- 97.71	110.53 +/- 6.69	0.06 +/- 0.08	126.33 +/- 48.75
logspiral	227.79 +/- 32.12	131.22 +/- 2.63	0.46 +/- 0.35	97.36 +/- 16.70
ars	126.20 +/- 73.24	81.24 +/- 22.17	0.18 +/- 0.18	98.66 +/- 38.39

EFFECT SIZES - NNN3 vs baselines (Cohen's d)

void

vs ars : food_eaten=+inf (very large), area_visited=+5.90 (very large), spiral_quality=+0.98 (large), mean_speed=+0.97 (large)
 vs crw : food_eaten=+inf (very large), area_visited=+5.83 (very large), spiral_quality=+0.97 (large), mean_speed=+0.97 (large)
 vs levy : food_eaten=+inf (very large), area_visited=+5.74 (very large), spiral_quality=+0.98 (large), mean_speed=+0.98 (large)
 vs logspiral : food_eaten=+inf (very large), area_visited=+4.06 (very large), spiral_quality=-0.22 (small), mean_speed=+0.98 (large)
 vs nnn7 : food_eaten=+inf (very large), area_visited=+1.61 (very large), spiral_quality=-1.90 (very large), mean_speed=+0.98 (large)
 vs stochastic : food_eaten=+inf (very large), area_visited=+11.05 (very large), spiral_quality=+0.93 (large), mean_speed=+0.93 (large)

dense_band

vs ars : food_eaten=+1.40 (very large), area_visited=+1.08 (large), spiral_quality=-0.48 (small), mean_speed=+0.48 (small)
 vs crw : food_eaten=+1.55 (very large), area_visited=+0.01 (negligible), spiral_quality=+0.48 (small), mean_speed=+0.48 (small)
 vs levy : food_eaten=+1.54 (very large), area_visited=+0.12 (negligible), spiral_quality=+0.51 (medium), mean_speed=+0.51 (medium)
 vs logspiral : food_eaten=+1.48 (very large), area_visited=-0.70 (medium), spiral_quality=-3.27 (very large), mean_speed=+0.48 (small)
 vs nnn7 : food_eaten=+0.78 (medium), area_visited=+0.19 (negligible), spiral_quality=-0.41 (small), mean_speed=-0.41 (small)
 vs stochastic : food_eaten=+1.82 (very large), area_visited=+2.13 (very large), spiral_quality=+0.38 (small), mean_speed=+0.38 (small)

dense_bands

vs ars : food_eaten=+4.77 (very large), area_visited=-0.53 (medium), spiral_quality=-0.79 (medium), mean_speed=-0.79 (medium)
 vs crw : food_eaten=+4.85 (very large), area_visited=-3.66 (very large), spiral_quality=+0.12 (negligible), mean_speed=+0.12 (negligible)
 vs levy : food_eaten=+4.06 (very large), area_visited=-3.23 (very large), spiral_quality=+0.17 (negligible), mean_speed=+0.17 (negligible)
 vs logspiral : food_eaten=+4.35 (very large), area_visited=-5.26 (very large), spiral_quality=-1.52 (very large), mean_speed=-1.52 (very large)
 vs nnn7 : food_eaten=+0.91 (large), area_visited=+1.61 (very large), spiral_quality=-0.47 (small), mean_speed=-0.47 (small)
 vs stochastic : food_eaten=+6.09 (very large), area_visited=+1.10 (large), spiral_quality=+0.06 (negligible), mean_speed=+0.06 (negligible)

EFFECT SIZES - NNN7 vs baselines (Cohen's d)

void

vs ars : food_eaten=+inf (very large), area_visited=+5.11 (very large), spiral_quality=+3.87 (very large), mean_speed=+3.87 (very large)
 vs crw : food_eaten=+inf (very large), area_visited=+5.01 (very large), spiral_quality=+3.84 (very large), mean_speed=+3.84 (very large)
 vs levy : food_eaten=+inf (very large), area_visited=+4.86 (very large), spiral_quality=+3.89 (very large), mean_speed=+3.89 (very large)
 vs logspiral : food_eaten=+inf (very large), area_visited=+2.73 (very large), spiral_quality=+1.04 (large), mean_speed=+1.04 (large)
 vs nnn3 : food_eaten=+inf (very large), area_visited=-1.61 (very large), spiral_quality=+1.90 (very large), mean_speed=+1.90 (very large)
 vs stochastic : food_eaten=+inf (very large), area_visited=+11.34 (very large), spiral_quality=+3.70 (very large), mean_speed=+3.70 (very large)

dense_band

vs ars : food_eaten=+0.87 (large), area_visited=+1.06 (large), spiral_quality=-0.03 (negligible), mean_speed=+1.
vs crw : food_eaten=+1.12 (large), area_visited=-0.30 (small), spiral_quality=+0.82 (large), mean_speed=+1.27 (v
vs levy : food_eaten=+1.11 (large), area_visited=-0.16 (negligible), spiral_quality=+0.85 (large), mean_speed=+1.
vs logspiral : food_eaten=+1.02 (large), area_visited=-1.24 (very large), spiral_quality=-2.33 (very large), mean_spee
vs nnn3 : food_eaten=-0.78 (medium), area_visited=-0.19 (negligible), spiral_quality=+0.41 (small), mean_speed=+0
vs stochastic : food_eaten=+1.59 (very large), area_visited=+2.45 (very large), spiral_quality=+0.74 (medium), mean_spe

dense_bands

vs ars : food_eaten=+2.43 (very large), area_visited=-1.84 (very large), spiral_quality=-0.30 (small), mean_spee
vs crw : food_eaten=+2.58 (very large), area_visited=-9.68 (very large), spiral_quality=+0.55 (medium), mean_spe
vs levy : food_eaten=+2.23 (very large), area_visited=-7.96 (very large), spiral_quality=+0.58 (medium), mean_spe
vs logspiral : food_eaten=+1.65 (very large), area_visited=-13.11 (very large), spiral_quality=-1.21 (very large), mea
vs nnn3 : food_eaten=-0.91 (large), area_visited=-1.61 (very large), spiral_quality=+0.47 (small), mean_speed=+1.
vs stochastic : food_eaten=+3.16 (very large), area_visited=-0.91 (large), spiral_quality=+0.48 (small), mean_speed=+1.
RANK ORDERING (best to worst per metric)

void

food_eaten : ars(0.00) > crw(0.00) > levy(0.00) > logspiral(0.00) > nnn3(0.00) > nnn7(0.00) > stochastic(0.00)
area_visited : nnn3(166.91) > nnn7(149.57) > logspiral(131.06) > crw(113.67) > ars(113.10) > levy(110.36) > stoch
spiral_quality : nnn7(0.67) > logspiral(0.33) > nnn3(0.25) > stochastic(0.08) > crw(0.07) > levy(0.07) > ars(0.07)
mean_speed : nnn7(138.39) > nnn3(133.29) > ars(130.54) > crw(127.71) > levy(123.72) > logspiral(88.84) > stocho

dense_band

food_eaten : nnn3(372.53) > nnn7(213.12) > ars(119.59) > logspiral(105.84) > crw(87.69) > levy(87.28) > stochas
area_visited : logspiral(133.09) > nnn3(114.48) > crw(114.08) > levy(111.29) > nnn7(108.06) > ars(83.12) > stocho
spiral_quality : logspiral(0.60) > ars(0.19) > nnn7(0.19) > nnn3(0.12) > stochastic(0.07) > crw(0.06) > levy(0.06)
mean_speed : nnn7(274.44) > logspiral(204.19) > nnn3(185.80) > levy(155.72) > crw(145.03) > ars(110.55) > stocho

dense_bands

food_eaten : nnn3(482.85) > nnn7(385.25) > logspiral(227.79) > levy(126.89) > ars(126.20) > crw(105.11) > stocho
area_visited : logspiral(131.22) > crw(113.62) > levy(110.53) > ars(81.24) > nnn3(71.02) > stochastic(57.45) > nn
spiral_quality : logspiral(0.46) > ars(0.18) > nnn7(0.13) > nnn3(0.07) > stochastic(0.07) > crw(0.06) > levy(0.06)
mean_speed : crw(130.87) > levy(126.33) > ars(98.66) > logspiral(97.36) > nnn7(82.71) > stochastic(63.13) > nnn

Table S13. External baseline comparison: descriptive statistics, Cohen's d effect sizes, and rank orderings across three environments.

Table S14: Formal Statistical Tests

Formal inferential statistics for all 95 validation checks. All directional tests are one-tailed in the predicted direction. Comparisons: Welch's t-test. Thresholds: one-sample t-test. Tolerance: TOST with stated margin. Ratios: bootstrap (10,000 resamples). Ordering: pairwise one-tailed t-tests. Holm-Bonferroni correction across all 95 tests. Obs = pattern observed in sample means. Inf = adj p < 0.05 after correction. Mann-Whitney U p-values available in the full table (Docs/results/stat/formal_tests_table.md).

#	Hyp	Description	Type	Adj p	d	d 95% CI	Obs	Inf
1	Ho	Hoa straightness >> Hob (ratio=4.0)	ratio	< 1e-200	2.19	[2.08, 2.31]	yes	yes
2	Ho	Hoa radius >> Hob (ratio=14.12)	ratio	< 1e-200	5.85	[5.65, 6.05]	yes	yes
3	Ho	Hob circle_fit > 0.3	threshold	< 1e-200	6.31	[6.03, 6.60]	yes	yes
4	Ho	Hoc circle_fit > 0.3	threshold	< 1e-200	6.38	[6.10, 6.67]	yes	yes
5	Ho	Hob radius ≈= Hoc radius	tolerance (T	< 1e-200	0.04	[-0.05, 0.13]	yes	yes
6	Ho	Hod speed ≈= 2x Hob (ratio=2.00)	ratio	< 1e-200	156677	[151822.12, 16	yes	yes
7	Ho	Hod radius ≈= Hob radius	tolerance (T	< 1e-200	0.26	[0.17, 0.34]	yes	yes
8	Ho	Hoe speed ≈= 4x Hob (ratio=4.01)	ratio	< 1e-200	294363	[285241.51, 30	yes	yes
9	Ho	Hoe radius ≈= Hob radius	tolerance (T	< 1e-50	0.82	[0.73, 0.91]	yes	yes
10	Ho	Hod radius ≈= Hoe radius	tolerance (T	< 1e-200	-0.57	[-0.66, -0.48]	yes	yes
11	Ho	Hob radius > Hof radius	comparison	< 1e-200	9.01	[8.72, 9.30]	yes	yes
12	Ho	Hof circle_fit > 0.3	threshold	< 1e-200	6.33	[6.04, 6.61]	yes	yes
13	Ho	Radius: Hob > Hof > Hog	ordering	< 1e-200	3.60	-	yes	yes
14	Ho	Hog circle_fit > 0.3	threshold	< 1e-200	6.59	[6.30, 6.89]	yes	yes
15	H1	H1a spiral_quality > 0.15	threshold	0.0275	0.08	[0.02, 0.14]	yes	yes
16	H1	H1a growth_rate > 0	threshold	< 1e-50	0.55	[0.48, 0.61]	yes	yes
17	H1	H1a straightness < 0.3	threshold	< 1e-200	-6.53	[-6.82, -6.23]	yes	yes
18	H1	H1b spiral >= H1a	comparison	2.61e-12	0.33	[0.24, 0.41]	yes	yes
19	H1	H1b area >= H1a	comparison	< 1e-50	1.49	[1.39, 1.59]	yes	yes
20	H1	H1c spiral >= H1b	comparison	3.24e-06	0.22	[0.13, 0.31]	yes	yes
21	H1	H1c area >= H1b	comparison	1.69e-40	0.62	[0.53, 0.71]	yes	yes
22	H1	Area: H1a <= H1b <= H1c	ordering	1.69e-40	0.62	-	yes	yes
23	H2	H2a spiral < 0.20	threshold	< 1e-200	-1.68	[-1.78, -1.59]	yes	yes
24	H2	H2b spiral < 0.20	threshold	< 1e-200	-1.45	[-1.54, -1.36]	yes	yes
25	H2	H2c spiral < 0.20	threshold	4.54e-46	-0.48	[-0.55, -0.42]	yes	yes
26	H2	H2d spiral < 0.20	threshold	< 1e-200	-1.73	[-1.83, -1.63]	yes	yes
27	H2	H2e spiral < 0.20 (opp w/o stagger	threshold	< 1e-200	-1.32	[-1.41, -1.24]	yes	yes
28	H2	H2f spiral < 0.20 (stagger w/o opp	threshold	< 1e-50	-1.08	[-1.16, -1.00]	yes	yes
29	H2	H2g spiral > 0.20 (positive ctrl)	threshold	< 1e-50	0.67	[0.60, 0.74]	yes	yes
30	H2	H2g spiral > 2x max(H2a-H2f) (rati	ratio	< 1e-200	1.48	[1.38, 1.58]	yes	yes
31	H2	H2g >> H2e (stagger matters)	comparison	< 1e-50	1.48	[1.38, 1.58]	yes	yes
32	H2	H2g >> H2f (opp matters)	comparison	< 1e-50	1.40	[1.30, 1.50]	yes	yes
33	H2	H2g area > H2b area	comparison	< 1e-200	14.21	[13.76, 14.66]	yes	yes
34	H3	H3a pre-food spiral > 0.12	threshold	< 1e-50	1.09	[1.01, 1.17]	yes	yes
35	H3	H3b pre-food spiral > 0.10	threshold	< 1e-200	1.49	[1.40, 1.58]	yes	yes
36	H3	H3c ctrl spiral < 0.10	threshold	1.23e-11	-0.25	[-0.32, -0.18]	yes	yes
37	H3	H3d ctrl spiral < 0.10	threshold	1.37e-05	-0.17	[-0.24, -0.10]	yes	yes
38	H3	H3a spiral > H3c ctrl	comparison	< 1e-50	1.54	[1.44, 1.65]	yes	yes
39	H3	H3b spiral > H3d ctrl	comparison	< 1e-200	1.81	[1.69, 1.92]	yes	yes
40	H3	H3b post-food revisit > 0.90	threshold	< 1e-200	1.87	[1.77, 1.98]	yes	yes
41	H3	H3a food > H3c ctrl	comparison	< 1e-200	1.97	[1.87, 2.08]	yes	yes
42	H3	H3b food > H3d ctrl	comparison	< 1e-50	1.21	[1.11, 1.30]	yes	yes
43	H4a	Area: H4a4>=H4a3>=H4a2>=H4a1	ordering	< 1e-200	4.96	-	yes	yes
44	H4a	H4a3 area > 5x H4a1 (ratio=7.18)	ratio	< 1e-200	9.20	[8.90, 9.50]	yes	yes
45	H4a	Speed: H4a5>=H4a4>=H4a3>=H4a2>=H4a	ordering	< 1e-50	0.73	-	yes	yes
46	H4a	H4a5 speed > 2x H4a1 (ratio=2.39)	ratio	< 1e-200	5.99	[5.78, 6.19]	yes	yes
47	H4a	H4a5 area > H4a4	comparison	< 1e-200	2.04	[1.93, 2.15]	yes	yes
48	H4a	H4a3 area > H4a7	comparison	< 1e-200	6.15	[5.94, 6.36]	yes	yes
49	H4a	H4a5 area > H4a7	comparison	< 1e-200	20.96	[20.30, 21.61]	yes	yes
50	H4a	H4a6 speed > H4a1	comparison	< 1e-200	2.50	[2.38, 2.62]	yes	yes
51	H4a	H4a2 area > H4a1	comparison	< 1e-200	6.00	[5.80, 6.21]	yes	yes
52	H4a	H4a4 area > H4a3	comparison	< 1e-200	4.96	[4.79, 5.14]	yes	yes
53	H4a	H4a4 speed > H4a3	comparison	< 1e-50	0.73	[0.64, 0.83]	yes	yes
54	H4a	H4a6/H4a3 area ratio 0.3-3.0	tolerance (T	< 1e-50	0.65	[0.56, 0.74]	yes	yes
55	H4b	H4b1 baseline spiral > 0.15	threshold	1.75e-27	0.36	[0.30, 0.42]	yes	yes
56	H4b	Spiral: 0%>=10%>=20%>=30%	ordering	0.0417	0.10	-	yes	yes
57	H4b	H4b4 30% spiral > 0.05	threshold	4.18e-25	0.34	[0.28, 0.41]	yes	yes
58	H4b	H4b2 retains > 50% baseline (ratio	ratio	0.0024	-0.45	[-0.54, -0.36]	yes	yes
59	H4b	Area: 0%>=10%>=20%>=30%	ordering	6.91e-17	0.39	-	yes	yes
60	H5	Food: H5a4>=H5a3>=H5a2>=H5a1	ordering	4.82e-41	0.62	-	yes	yes
61	H5	H5a4 food > 1.5x H5a1 (ratio=1.64)	ratio	< 1e-200	1.79	[1.69, 1.89]	yes	yes

#	Hyp	Description	Type	Adj p	d	d 95% CI	Obs	Inf
62	H5	H5a4 food > H5a3	comparison	1.67e-46	0.67	[0.58, 0.76]	yes	yes
63	H5	H5a3 food > H5a2	comparison	4.82e-41	0.62	[0.53, 0.71]	yes	yes
64	H5	H5a2 food > H5a1	comparison	< 1e-50	0.86	[0.77, 0.95]	yes	yes
65	H5	H5a4 food > 10	threshold	< 1e-200	3.86	[3.68, 4.04]	yes	yes
66	H5	Area: H5a4>=H5a3>=H5a2>=H5a1	ordering	< 1e-50	1.27	-	yes	yes
67	H6	Food: H6a1>=H6a2>=H6a3>=H6a4>=H6a5	ordering	0.1025	0.08	-	yes	no
68	H6	H6a1 food > 1.3x H6a5 (ratio=1.41)	ratio	< 1e-200	1.33	[1.23, 1.43]	yes	yes
69	H6	H6a1 food > H6a2	comparison	0.1025	0.08	[-0.01, 0.17]	yes	no
70	H6	H6a2 food > H6a3	comparison	2.81e-26	0.49	[0.40, 0.58]	yes	yes
71	H6	H6a1 food > 10	threshold	< 1e-200	3.84	[3.66, 4.01]	yes	yes
72	H6	Area: H6a1>...>=H6a5	ordering	0.4817	0.00	-	yes	no
73	H7	H7a1 D_ROT=0 spiral > 0.20	threshold	< 1e-200	3.36	[3.20, 3.52]	yes	yes
74	H7	H7a1 D_ROT=0 spiral > 0.50 (high q	threshold	< 1e-50	1.13	[1.05, 1.21]	yes	yes
75	H7	H7a4 D_ROT=0.1 spiral > 0.10	threshold	< 1e-50	0.77	[0.70, 0.84]	yes	yes
76	H7	H7a6 D_ROT=0.5 spiral > 0.05	threshold	< 1e-50	0.89	[0.81, 0.96]	yes	yes
77	H7	H7a6 area > H7a1 area	comparison	< 1e-200	9.12	[8.83, 9.42]	yes	yes
78	H8	H8a5 spiral > H8a1	comparison	< 1e-200	2.10	[1.99, 2.21]	yes	yes
79	H8	H8a5 spiral > 1.3x H8a1 (ratio=2.6	ratio	< 1e-200	2.10	[1.99, 2.21]	yes	yes
80	H8	H8a5 spiral > 0.50 (high)	threshold	< 1e-200	1.29	[1.20, 1.37]	yes	yes
81	H8	H8a5 spiral > H8a1 (endpoints)	comparison	< 1e-200	2.10	[1.99, 2.21]	yes	yes
82	H8	H8a1 weakest spiral > 0.05	threshold	< 1e-50	0.88	[0.81, 0.95]	yes	yes
83	H9	NEG-H0 radius > 200	threshold	< 1e-50	0.55	[0.49, 0.62]	yes	yes
84	H9	NEG-H1 spiral < 0.10	threshold	5.18e-24	-0.33	[-0.40, -0.27]	yes	yes
85	H9	NEG-H2 spiral < 0.20	threshold	< 1e-50	-0.65	[-0.72, -0.58]	yes	yes
86	H9	NEG-H3 pre_food_spiral < 0.10	threshold	< 1e-200	-1.64	[-1.75, -1.53]	yes	yes
87	H9	NEG-H3 food < 100	threshold	< 1e-50	-0.90	[-0.98, -0.83]	yes	yes
88	H9	NEG-H4a spiral < 0.20	threshold	< 1e-50	-0.90	[-0.98, -0.83]	yes	yes
89	H9	NEG-H4b spiral < 0.10	threshold	1.76e-39	-0.44	[-0.51, -0.38]	yes	yes
90	H9	NEG-H5 food < 1	threshold	< 1e-200	>10	[nan, inf]	yes	yes
91	H9	NEG-H6 food < 1	threshold	< 1e-200	>10	[nan, inf]	yes	yes
92	H9	NEG-H7 area < 15	threshold	< 1e-200	-4.19	[-4.38, -3.99]	yes	yes
93	H9	NEG-H7 circle_fit > 0.95	threshold	< 1e-200	25242.	[24136.01, 263	yes	yes
94	H9	NEG-H8a spiral < 0.10	threshold	2.05e-13	-0.24	[-0.31, -0.18]	yes	yes
95	H9	NEG-H8b spiral < 0.10	threshold	< 1e-50	-0.69	[-0.76, -0.62]	yes	yes

Table S14. Formal statistical tests for all 95 validation checks. 95/95 observed in predicted direction; 92/95 inferentially supported (3 non-inferred are H6 ordering checks with $|d| < 0.10$).